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Charged hadron distributions in Z^0 tagged jets

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Abstract

We present an analysis of charged hadron-in-jet distributions for jets that are nearly back-to-back with a measured Z boson. The measurement is made in the 2012 $\sqrt{s}$ = 8 TeV data set, corresponding to a total integrated luminosity of approximately 2 fb^{-1}. The longitudinal, transverse, and radial distributions of charged hadrons in the jet are studied as a function of the jet p_{T}. We also explore new observables related to hadronization and jet substructure. Where calculations are possible for a particular observable, the results are compared to NLO perturbative QCD calculations using soft-collinear effective theory and to predictions from several event generators.

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1 Introduction

One of the outstanding questions in quantum chromodynamics (QCD) research is understanding how partons combine to form hadrons. Since this is inherently a nonperturbative process, first principles perturbative QCD (pQCD) calculations cannot predict the distributions of partons within hadrons. Therefore, global fits are necessary to constrain the nonperturbative description of partons within hadrons and require a wide variety of data and measurements in unpolarized and polarized high energy collisions. For example, there has been significant effort in the last two decades in understanding the initial-state transverse momentum and transverse spatial distributions of partons within the proton (for a review see e.g. Refs. [1, 2]). However, the ideas and concepts behind hadronization, the process by which partons nonperturbatively form bound state hadrons after undergoing a hard scatter, are much less developed than their initial-state counterparts. Thus, measurements of final-state hadrons in relation to the original final-state partons are required to better understand the process of hadronization.

Currently, our knowledge of fragmentation functions and hadronization comes mostly from e^+e^- collisions; this is because there are no initial-state partons and thus the annihilation process provides a clean channel to study hadrons in the final-state [3, 4]. Examples of measurements with this channel can be found in the references of global fragmentation function fits, e.g. Refs. [3, 4]. However, these collisions are typically at small center-of-mass energies where a partonic picture within pQCD can become more complicated. Thus, high energy hadron colliders provide a unique opportunity to study final-state hadrons since the momentum transfer of high p_{T} processes is large, and thus a picture relating the partonic to hadronic degrees of freedom can be applied. To make this relation, jets are typically used to study hadron-in-jet distributions. For example, the hadron-in-jet distributions can be studied with next-to-leading order (NLO) pQCD calculations [5] or within soft-collinear effective theory (SCET) [6]. Typically the fragmentation functions are studied as a function of the longitudinal momentum fraction of the hadron with respect to the parton, as this can be determined exactly in e^+e^- collisions. The reconstruction of the full jet additionally allows for transverse distributions to be studied. These are sensitive to the transverse-momentum-dependent (TMD) fragmentation functions, dependent on both the longitudinal momentum fraction z of the hadron with respect to the jet axis as well as the hadron transverse momentum j_T with respect to the jet axis. In this way nonperturbative hadronization studies can be performed multidimensionally, which extends previous measurements only sensitive to the longitudinal dynamics of final-state hadrons. Multidimensional phenomenological calculations have steadily progressed over the past decade, showing the need for more data with which to constrain nonperturbative functions (see e.g. Refs. [7, 8] and Refs. within).

The LHCb experiment provides a unique opportunity to study distributions of hadrons-in-jets. As an experiment built for precision tracking, the LHCb apparatus provides excellent track finding capability within jets. Additionally, as the only experiment at a high-energy hadronic collider in the world with excellent particle identification, the experiment offers a wide variety of possibilities for studying identified particles within jets and/or correlations of identified particles within jets. Further, with explicit charm and bottom quark tagging, LHCb provides a rare opportunity to explicitly tag flavors of partons (udsg, c, b) and then subsequently relate them to their identified hadron constituents in the corresponding jet. With these considerations, LHCb is one of the only

active experiments in the world at a hadronic collider suited for robust and comprehensive measurements of hadrons within jets, where each hadron can be explicitly identified multidifferentially and with particle identification.

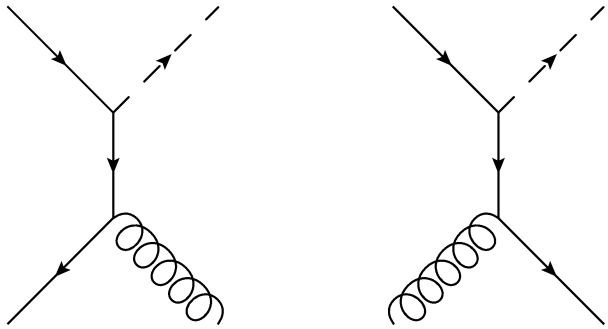

Figure 1: The leading order partonic processes that contribute to Z-jet production in QCD. The left shows the $q\bar{q}$ annihilation process, while the right shows the QCD Compton scattering process.

LHCb has already measured forward Z-jet[1] cross sections [9, 10], and this channel offers several advantages to studying hadronization in a jet. Firstly, the Z-jet process has incredibly high signal-to-background, making it an extremely clean process to measure. Previous LHCb analyses have measured the background fraction of Z bosons to be approximately 99% [11, 12], while the background from fake jets is usually less than 1% [9, 10].Secondly, Z-jet production is frequently produced by the QCD Compton scattering process, where a quark absorbs a gluon and emits a Z boson. Thus, the Z-jet channel provides additional information compared to the inclusive jet channel because the jets are predominantly light quark jets. The two leading order partonic processes for Z-jet production are shown in Fig. 1. Therefore, it provides the opportunity to study a high fraction of (light) quark fragmentation functions compared to gluon fragmentation functions. It is also worth mentioning that LHCb has measured both the Z-jet and $Z + b$ jet cross section at $\sqrt{s} = 7$ TeV [9, 13]; the ratios of these total cross sections is approximately 2%, which, modulo the charm contribution, provides some potential limits on heavy quark vs. light quark contributions in this measurement. Finally, measuring a jet in correlation with a Z boson allows for a higher degree of accuracy to the partonic kinematics. This is because a complete $2 \to 2$ hard scattering process can be inferred from the process, where the two initial-state partons lead to two final-state objects (in this case, the Z and the jet). This allows theorists to calculate the longitudinal momentum fraction x of the incoming partons, which provides a better understanding the underlying partonic process. Thus, the process gives a more complete picture of the event at the partonic level, and provides an opportunity to connect quantitative partonic degrees of freedom to the quantitative final-state hadron degrees of freedom.

This note presents an analysis of charged hadrons in jets tagged in correlation with a Z boson using data collected in 2012 by LHCb at $\sqrt{s} = 8$ TeV. The fragmentation distributions of the charged hadrons with respect to the jet are studied, defined as

[1]The notation Z additionally includes contributions from virtual photon production and its interference with Z boson production, γ^*/Z.

$$z \equiv \frac{\mathbf{p}_{jet} \cdot \mathbf{p}_{chg}}{|\mathbf{p}_{jet}|^2}, \tag{1}$$

$$j_T \equiv \frac{|\mathbf{p}_{jet} \times \mathbf{p}_{chg}|}{|\mathbf{p}_{jet}|}, \tag{2}$$

$$r = \Delta R(chg, jet) \equiv \sqrt{(\phi_{chg} - \phi_{jet})^2 + (y_{chg} - y_{jet})^2}, \tag{3}$$

where $\mathbf{p}$ is the momentum vector, ϕ is the azimuthal angle, y is the rapidity, and the subscript for each refers to either the jet or the charged hadron. These distributions quantify the longitudinal momentum fraction, transverse momentum, and radial distribution of hadrons within the jet, respectively. Measurements[2] are made as a function of the jet p_{T} in the LHCb fiducial space covered by $2 < \eta < 4.5$ and for jet $p_{\mathrm{T}} > 20$ GeV with jet R = 0.5. The normalization of the distributions is determined by the total number of Z-jet in a given p_{T} bin, so the final observables are, for example in the case of z,

$$\frac{1}{N_{Z+jet}} \frac{dN}{dz}. \tag{4}$$

Equation 4 would then give the number of hadrons in a given z bin per jet in Z tagged events. We also explore the possibility of several other observables in this note, as one goal of this analysis is to characterize hadron distributions in jets as multidifferentially as the data will statistically and systematically allow. One can also study, for example, the momentum and p_{T} distributions of tracks within jets, which are similar to z and j_T. Similarly to r we can define the observable θ^{track} as

$$\theta^{track} = \mathrm{acos}\left(\frac{\mathbf{p}_{jet} \cdot \mathbf{p}_{had}}{|\mathbf{p}_{jet}||\mathbf{p}_{had}|}\right), \tag{5}$$

which defines the angle between the hadron and jet momentum vectors.

This note is organized in the following manner. Section 2 introduces the method used to measure the distributions. Section 3 describes the event and selection criterion for the Z, jet, and tracks within the jet. Section 4 discusses the efficiencies for tracks and jets and their application to the total multiplicities. Section 5 discusses the unfolding procedure, and Section 6 shows the systematic uncertainties assigned. There have been no measurements made of these observables at the Relativistic Heavy Ion Collider (RHIC), while at the LHC a measurement of these observables in inclusive jet production at central rapidity has been made [14–16]. Several measurements have been made of z and p_T^{track} from ATLAS and CMS; however, these measurements are largely focused on understanding QCD effects in relation to either cold nuclear matter effects and/or hot nuclear matter effects from the quark-gluon plasma [17–19]. The measurements presented in this note constitute the first measurements of charged hadron distributions in jets at LHCb, and thus access a unique forward kinematic phase space which the other LHC experiments typically cannot reach. Additionally, these are the first measurements of charged hadrons-in-jets correlated with a Z boson, rather than inclusive jets, from the LHC.

[2]This note uses natural units, where the speed of light c and the reduced Planck constant $\hbar$ are set to unity.

2 Analysis Strategy

The methods used to determine the observables defined in Section 1 are similar to previous analyses published by the LHCb collaboration [20] and the ATLAS collaboration [14, 18]. We are interested in the charged hadron multiplicities normalized per Z-jet, which means the desired distributions are, for example,

$$\frac{1}{N_{Z+jet}}\frac{dN}{dz}, \frac{1}{N_{Z+jet}}\frac{dN}{dj_T}, \frac{1}{N_{Z+jet}}\frac{dN}{dr}. \tag{6}$$

Therefore, in order to measure these distributions, it is necessary to measure both the true Z-jet spectrum to determine the normalization appropriately as well as the true charged hadron multiplicities. Due to the large jet energy and jet angle resolutions, the charged hadron distributions relative to the jet must be unfolded. The unfolding procedures will be discussed in more depth in Section 5; for now it suffices to give a general definition of the overall method. To determine the number of true jets in a given jet p_{T} bin an unfolding procedure must also be used; however, Section 4 will show that a bin-by-bin correction method can be used which provides equivalent results to a full unfold, similarly to Ref. [21]. The measured distributions are determined with

$$\frac{1}{N^{meas}_{Z+jet}}\frac{dN^{meas}_{track}}{dz} = \frac{\epsilon_\mu \epsilon_{jet}}{U_{jet} \times N^{reco}_{Z+jet}} \times \frac{1-\epsilon^{gained}_{track}}{\epsilon^{in\ jet}_{track}} \times \frac{dN^{reco}_{track}}{dz}. \tag{7}$$

This applies equally to any other observable, e.g. j_T or r, that one would like to define. In Eq. 7, N^{reco}_{Z+jet} is the number of reconstructed Z-jets in a given jet p_{T} bin, ϵ_μ is the total efficiency associated with muon, and thus Z boson, reconstruction, ϵ_{jet} is the jet efficiency, U_{jet} is an unfolding factor accounting for the jet p_{T} resolution, $\epsilon^{gained}_{track}$ is the efficiency with which tracks are misreconstructed inside of the truth jet cone, $\epsilon^{in\ jet}_{track}$ is the efficiency with which tracks are reconstructed inside of the truth jet cone, and dN^{reco}_{track}/dz are the number of charged hadrons in a given z bin divided by the bin width. The truth jet cone is defined as the jet reconstructed from stable MC particles. The track-in-jet efficiencies are constructed such that they correct the number of tracks back to the total number of tracks regardless of whether or not the jet was measured. Each of these factors will be discussed in more detail in Section 4; the purpose here is just to introduce the overall method.

Since $N^{meas}_{Z\text{-jet}}$ is simply a normalization factor, it can be determined separately from the unfolding of the charged hadron distributions. Thus, the factor $\frac{\epsilon_\mu \times \epsilon_{jet}}{U_{jet} \times N^{reco}_{jet}}$ can be determined for a given jet p_{T} bin. Note that in this analysis, the p_{T} of the Z boson is integrated over to acquire adequate statistical precision to measure the distributions as a function of jet p_{T}. The efficiency corrected charged hadron distributions can then be unfolded to determine the true charged hadron multiplicities, which in turn gives the final distributions after they are normalized. The integral of the distributions, when normalized in this way, then corresponds to the average multiplicity of the jets.

We test this method with the efficiencies defined as discussed in Section 4 and perform a closure test with MC. The same analysis that is used to determine the distributions in data is applied to the reconstructed MC, and compared to the truth MC. To perform this closure test on independent sets of MC data, the efficiencies and response matrices are determined with the magnet down data, and applied to the magnet up data. Figure 2 shows the

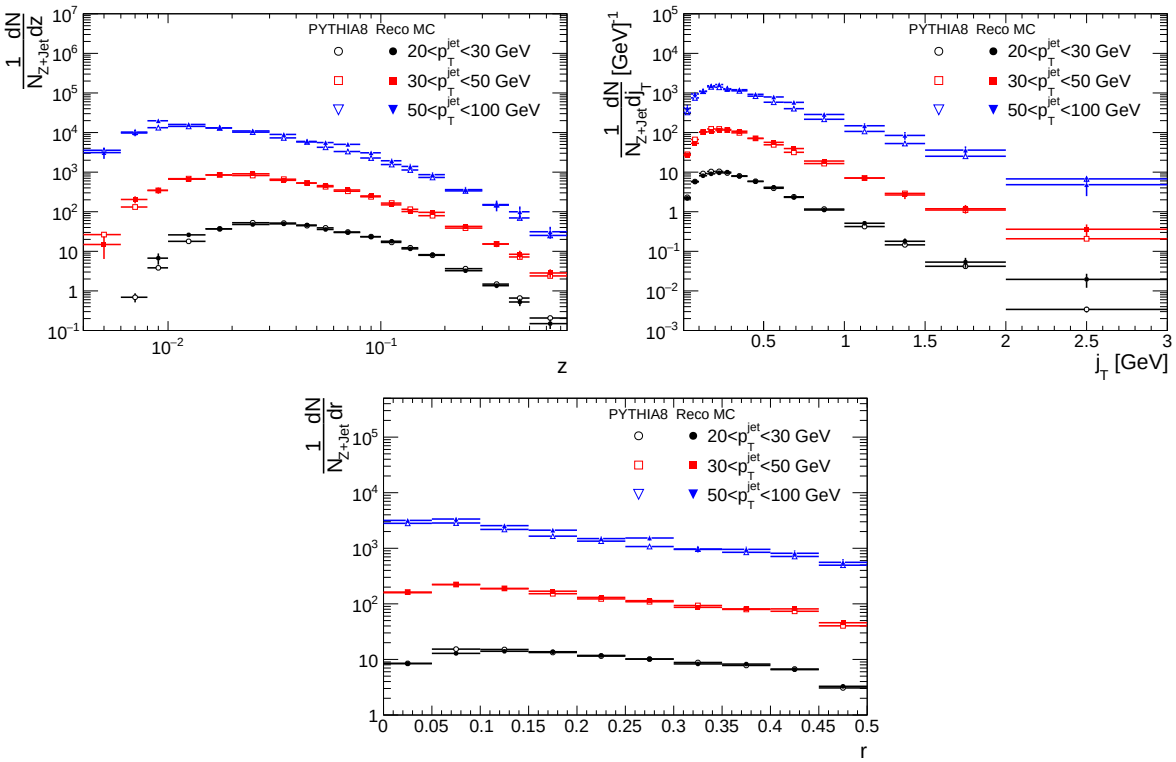

Figure 2: The truth MC and the efficiency corrected and unfolded reconstructed MC distributions are compared for several hadron-in-jet observables, using the method described above and the efficiencies as defined in Section 4. The distributions are scaled by an arbitrary factor of 10 for visibility sake. The uncertainties shown are from the statistical precision of the MC.

results of the closure test compared with the PYTHIA truth MC. The agreement between the truth distributions and the efficiency corrected unfolded reconstructed distributions is good, with the ratios shown in Fig. 3 all being consistent with unity.

3 Selection Criteria

Many of the selection criteria and analysis procedures are taken from the already established Z-jet cross section analyses performed in the 2011 and 2012 $\sqrt{s} = 7$ and 8 TeV data; the publications are found in Refs. [9, 10] and the corresponding internal LHCb analysis notes are found in Refs. [21, 22]. Since the 2012 data has a larger integrated luminosity, this data set was chosen to analyze; the total integrated luminosity is of more importance than the center-of-mass energy for this analysis since Z-jet production is a relatively rare process. Additionally, the Z-jet process has been thoroughly studied in the Run 1 data sets. Thus, the analysis is based on approximately 2 fb^{-1} of p+p data collected in the 2012 period. This corresponds to the stripping 20 data and MC samples that were produced with the Reco 14 production. The bookkeeping paths for the MC and data can be found in Appendix A.

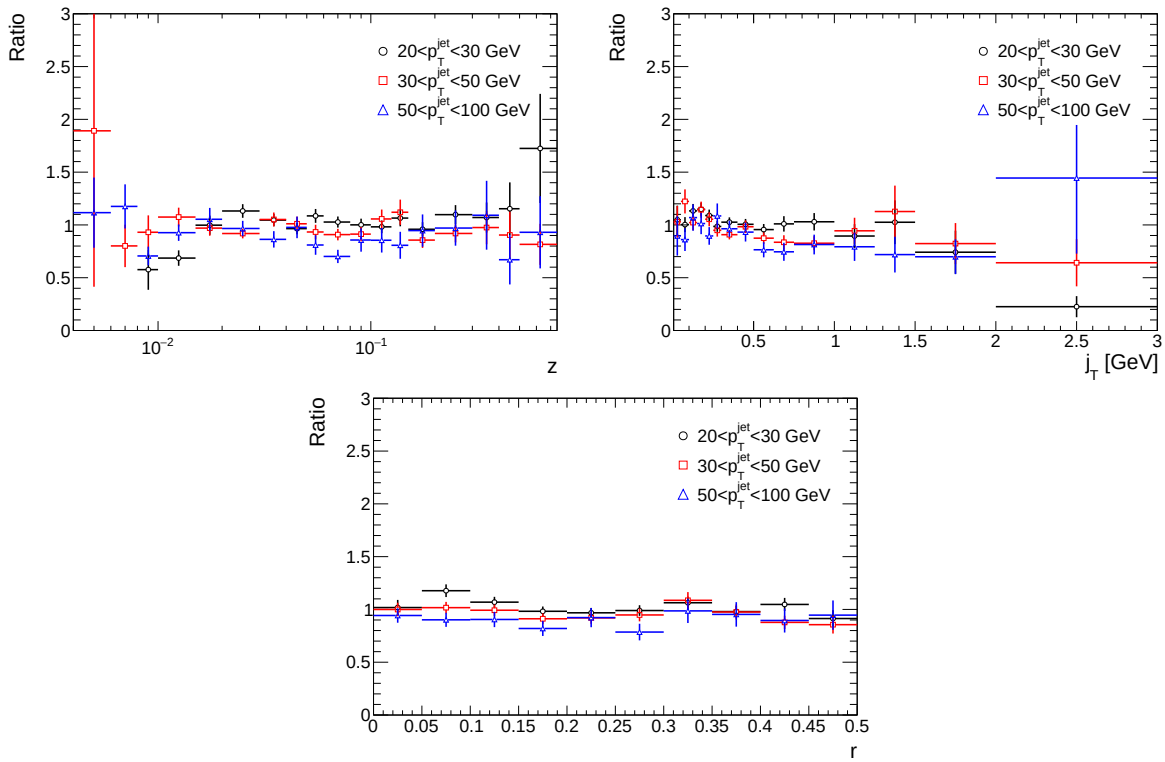

Figure 3: The ratio of the truth MC to the efficiency corrected and unfolded reconstructed MC using the method described above and the efficiencies as described in Section 4. The ratios are consistent with unity within the statistical uncertainties of the MC.

3.1 Code Location

A GitLab repository with the TupleTools used for this analysis can be found at the following link. An automated chain of analysis macros can be found in the repository at this link. Output tuples from the ganga grid jobs can be found on `eos` at the path `/eos/lhcb/wg/QEE/LHCb-ANA-2019-003`.

3.2 Event

There are several event level criterion that are placed on this analysis. Firstly, the event must have exactly one reconstructed primary vertex (PV). This criterion is placed similarly to other track analyses, for example the J/ψ-in-jet analysis [23], to ensure that the tracks being associated with the jet correspond to the same PV as the Z boson and the jet. This also reduces the fake jet rate and improves the jet resolution since neutral particles are then known to come from the single PV in the event. This is essential as we want to ensure that the event selection is selecting a $2 \to 2$ process, so that the partonic quantities correspond as closely as possible to the Z-jet quantities. The event is also required to be TOS on the same trigger line as the Z-jet cross section [21, 22]. At least one decay muon from the Z boson is required to be TOS all the way through the `L0Muon`, `Hlt1SingleMuonHighPT`, and `Hlt2SingleMuonHighPT` trigger lines. This provides a higher trigger efficiency, and thus more statistics, than the `L0DiMuon` line. The fiducial volume of the measurement will be discussed in the following sections outlining the Z, jet, and track requirements.

3.3 Z Boson

Standard quality cuts are applied to the muons and resulting Z boson to select candidate events. The reconstructed muons must satisfy the following criteria:

- $p_T > 20$ GeV
- $2 < \eta < 4.5$
- The relative uncertainty on the momentum of each muon is less than 10% - $\sigma_p/|p| < 0.1$
- The probability χ^2 of the associated track fit is required to be larger than 1%
- The muon is required to pass the `ISMUON` requirement

From these muon quality criterion, the resulting Z boson is required to have an invariant mass $60 < M_{\mu\mu} < 120$ GeV. Previous Z boson analyses from LHCb have shown that these cuts result in an approximately 99% pure sample of Z bosons [21,22,24,25]. The inclusive Z invariant mass distribution, in addition to the Z invariant mass distribution tagged with a jet, can be seen in Fig. 4. This shows that the presence of the jet does not introduce additional background into the Z mass distribution. For Z bosons tagged with an away-side jet, the p_{T} distribution of the Z is known to be broader than that of inclusive Z production due to the approximate p_{T} balance of the Z-jet. Figure 5 shows the p_{T} distribution for the inclusive Z and Z-jet channel in data, where the jet is selected with the criteria described below. In the subsequent analysis, the Z p_{T} is integrated over so that sufficient statistics can be acquired to measure the charged hadron-in-jet distributions as a function of the jet p_{T}. However, it is potentially useful to also know the range of the Z p_{T} distribution used in this analysis as displayed in Fig. 5.

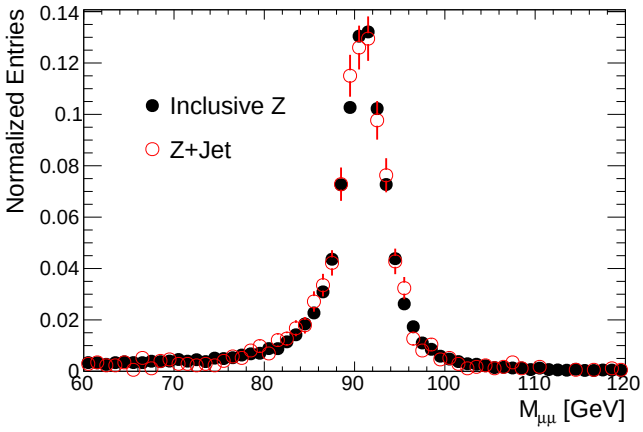

Figure 4: The inclusive Z invariant mass distribution and the Z invariant mass distribution in the presence of a jet are shown, where the event, Z, and jet selection criteria are described in the text.

We cross checked that the Z boson quantities matched in data and simulation, as was shown in the Z-jet cross section analysis in Ref. [21]. Figure 6 shows the inclusive Z boson p_{T} and rapidity distributions in both simulation and data. The histograms show that the simulation appropriately models the kinematics associated with reconstructing the Z boson in simulation.

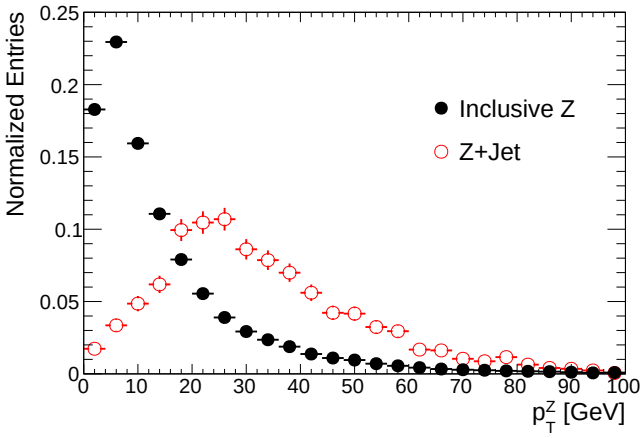

Figure 5: The Z p_{T} distribution is shown as measured in data for inclusive Z production and where the Z is produced in conjunction with a jet which passes the selection criterion detailed below. The Z p_{T} distribution for Z-jet production is pushed out to larger p_{T} due to the approximate p_{T} balance of the Z and high p_{T} jet. Note that, despite the y-axis being on a linear scale, there are no measured counts at Z p_{T} >100 GeV for Z bosons tagged with a jet in the data.

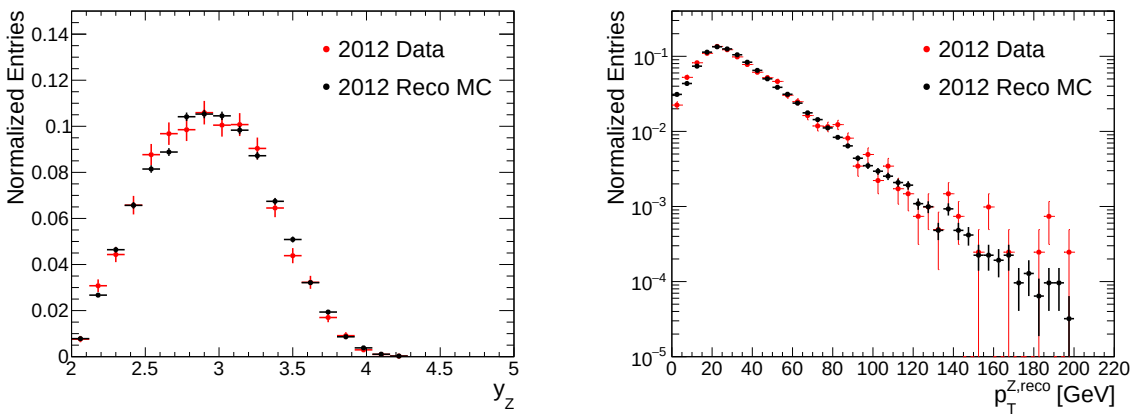

Figure 6: The inclusive Z boson rapidity and p_{T} distributions are shown in both reconstructed data and reconstructed MC. The distributions are consistent within uncertainties.

3.4 Jets

The jets are reconstructed using the standard LHCb jet reconstruction and jet identification requirements [26], which select high quality $R = 0.5$ jets with a very low fake rate. The jets are required to have p_{T} >15 GeV and to lie in the pseudorapidity range $2.5 < \eta < 4$. However, jets with $15 < p_{\mathrm{T}} < 20$ GeV are only analyzed for the purposes of the unfolding, which reduces the systematic uncertainty in the true jet p_{T} spectrum similarly to Ref. [23]. Thus, we only report measurements with jet $p_{\mathrm{T}} > 20$ GeV. In previous analyses jets have been collected within a pseudorapidity range of $2.2 < \eta < 4.2$; however, the reduced pseudorapidity range in this analysis is to ensure that the entire radius of the jet falls within the nominal LHCb track acceptance of $2 < \eta < 4.5$. Since the focus of this analysis is tracks within jets, the entire jet cone should lie within the LHCb acceptance so that there is not a bias to the number of tracks in a given jet due to some fraction of the jet

lying outside the fiducial volume of the detector. A requirement is also placed that the jet can not contain one of the final-state muons from the Z boson decay within its jet cone; in other words, $\Delta R > 0.5$, where ΔR is the radial distance in $\eta - \phi$ space between the jet and each final-state muon. Since one of the goals of this analysis is to also tag a high fraction of quark jets, it is important to identify events where there is a $2 \to 2$ scattering configuration. To tag the away-side jet in the scattering process, the Z-jet pair is also required to satisfy $\Delta\phi > 7\pi/8$, where $\Delta\phi = |\phi_Z - \phi_{jet}|$. These requirements, in addition to the PV requirement, reduce the fake jet rate to approximately 0.3%; this fake jet rate is discussed in more detail in the systematic uncertainty section. Examples of the jet p_{T}, η, and ϕ response can be found in Figs. 7 and 8, where the response is determined with simulation by matching truth MC jets to reconstructed jets in ΔR space similarly to Refs. [21, 22]. This requirement matches truth MC jets to reconstructed jets as well as in the previous analyses, where approximately 95% of reconstructed jets have a truth jet matched within $\Delta R < 0.1$. Figure 9 also shows the distribution of jet momentum after the selection criterion are applied, which is potentially relevant for understanding the range of z which can be accessed in this measurement.

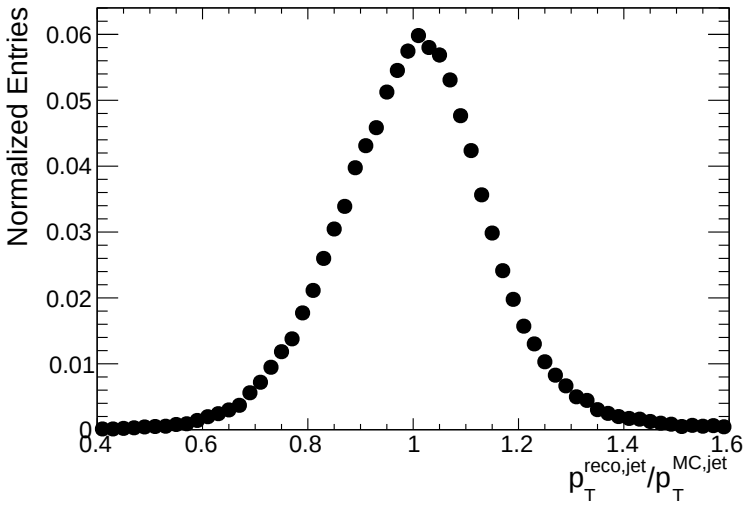

Figure 7: The jet p_{T} response, as determined in simulation, is shown after jet energy corrections are applied.

We also confirm the result of the Z-jet cross section analysis in that the data and reconstructed simulation match for the jet level quantities. Figure 10 shows that the data and reconstructed simulation match for the jet level quantities. Similar results have been found at both 7 and 8 TeV [21, 22].

3.5 Tracks-In-Jets

The charged hadron selection criterion within the jets is largely determined within the particle flow algorithm used to construct the jets [27]. Thus, these tracks have already been identified as jet daughters by the particle flow algorithm and are used to construct the jets. However, we apply several additional cuts here to ensure that the tracks are well reconstructed and have good efficiency within the LHCb acceptance. The charged hadrons are additionally required to satisfy the following criteria:

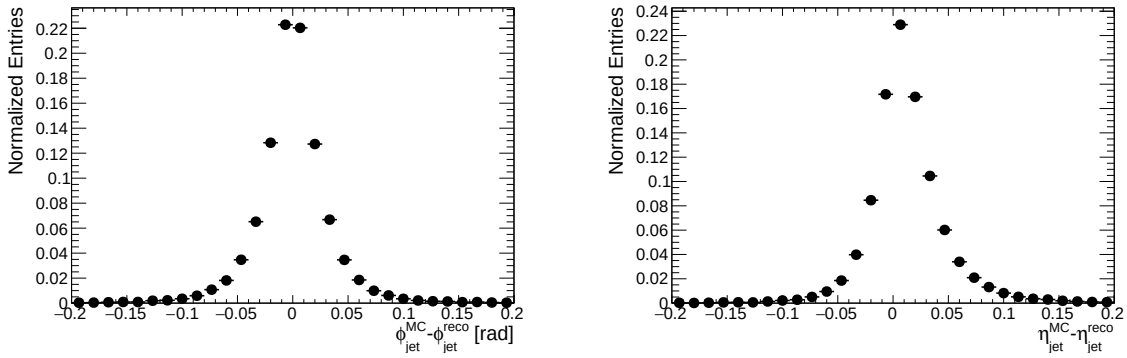

Figure 8: The jet ϕ (left) and η (right) response are shown for the jet selection criteria described in the text.

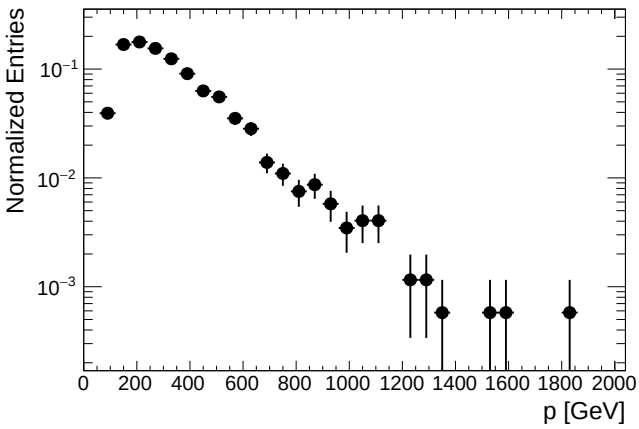

Figure 9: The total momentum of jets after the selection criteria are applied is shown.

- $4 < p < 1000$ GeV
- $p_{\mathrm{T}} > 0.25$ GeV
- Track χ^2/NDF<3

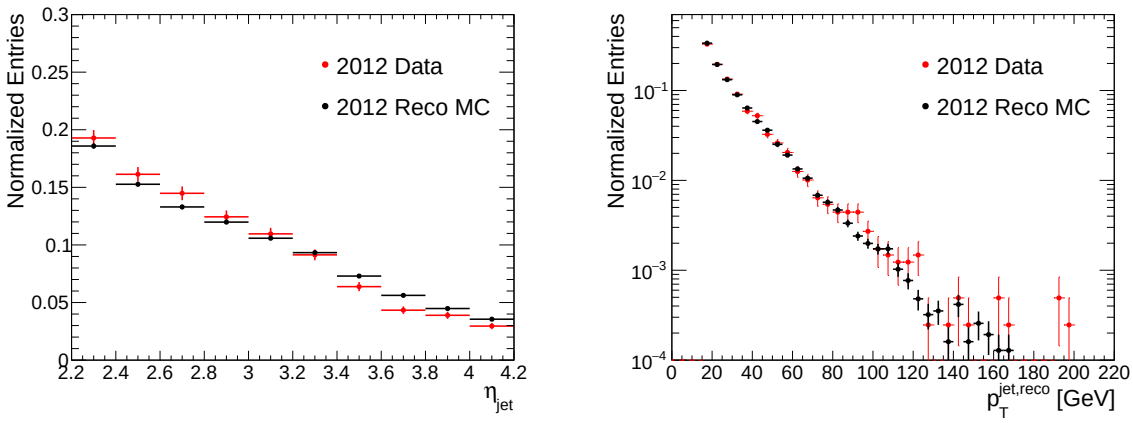

Figure 10: The jet pseudorapidity and p_{T} distributions are shown in both reconstructed data and reconstructed MC. The distributions are consistent within uncertainties.

- The track particle flow type (PFType) is required to be 2, which selects charged hadrons as determined by the particle flow algorithm

- ΔR <0.5, where ΔR is the distance in (η, ϕ) space between the track and the jet, and 0.5 is the jet cone size

Note that the requirement of $\Delta R_{track-jet} < 0.5$, in conjunction with the jet η requirement of $2.5 < \eta_{jet} < 4$, requires that the tracks be within the nominal LHCb acceptance of $2 < \eta_{track} < 4.5$. This requirement is also placed as there are instances where charged hadrons can fall at $\Delta R > R_{jet}$; however, they occur a very small fraction of the time. This occurs when the event topology corresponds to many lower p_T particles within the same area of phase space; in this case, the anti-k$_T$ algorithm takes the union of two $R = R_{jet}$ radius jets, which can allow some of the jet constituents to fall outside of the final jet cone. For a complete description of the various jet shapes that the anti-k_T jet algorithm constructs, see Section 2.1 of Ref. [28]. Figures 11, 12, and 13 show the hadron p_{T}, ϕ, and η resolution within jets as determined from the MC in several bins of hadron p_{T}. The figures show that the resolution for hadrons within jets, as a function of these three variables, is similar to the nominal resolution of tracks within LHCb.

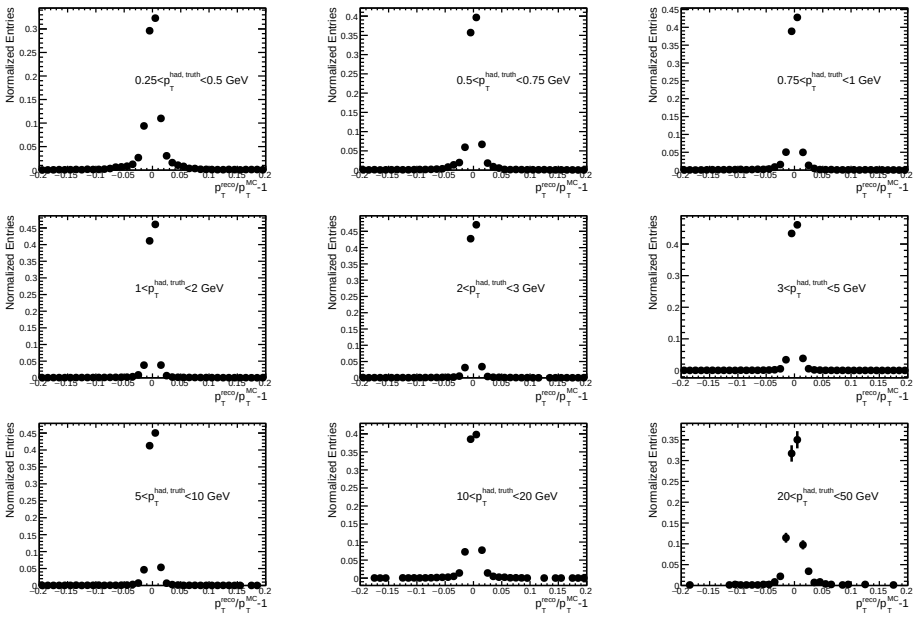

Figure 11: The hadron p_{T} resolution for hadrons within jets is shown in several hadron p_{T} bins.

3.6 Summary of Cuts

The following table shows a summary of the cuts placed to define the acceptance at truth level for the Z bosons, jets, and hadrons.

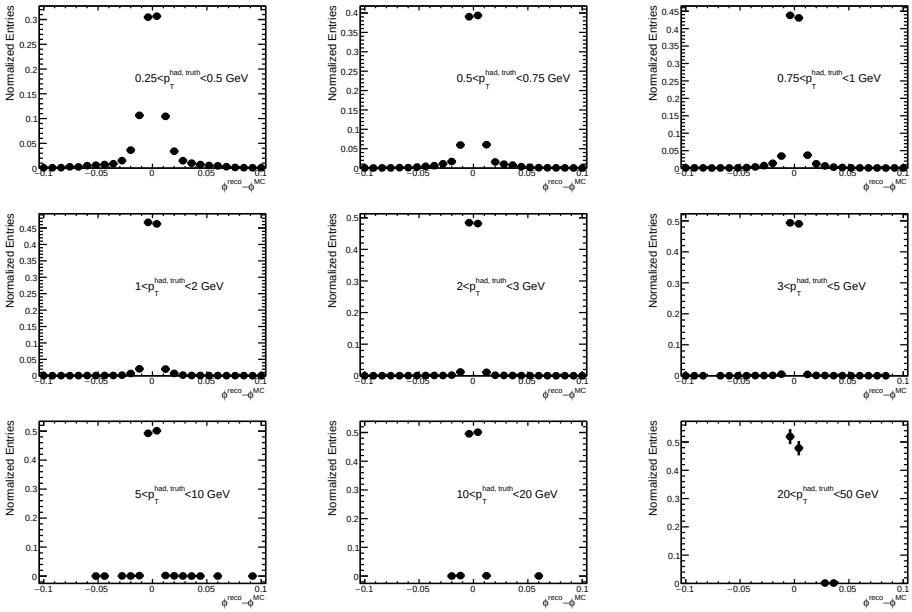

Figure 12: The hadron ϕ resolution for hadrons within jets is shown in several hadron p_{T} bins.

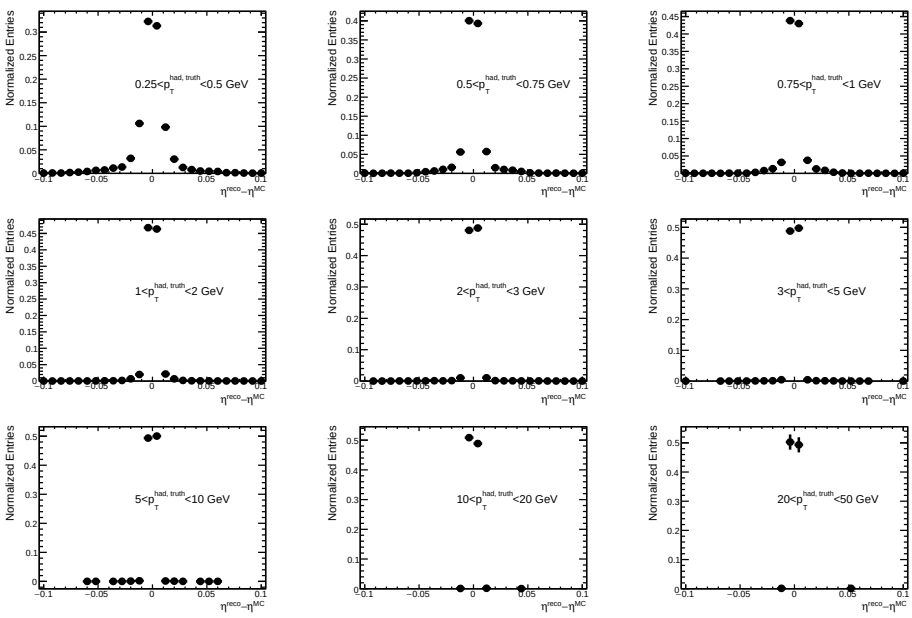

Figure 13: The hadron η resolution for hadrons within jets is shown in several hadron p_{T} bins.

Z Boson	Jet	Charged Hadrons
$0 < p_T^Z < 100$ GeV	$20 < p_{\mathrm{T}} < 100$ GeV	$p_{\mathrm{T}} > 0.250$ GeV
$2 < \eta_Z < 4.5$	$2.5 < \eta_{jet} < 4$	$4 < p < 1000$ GeV
$60 < M_{\mu\mu} < 120$ GeV	$\Delta\phi_{Z+jet} > 7\pi/8$	$\Delta R_{jet-track} < 0.5$

Table 1: The cuts applied to define the acceptance at truth level for the Z bosons.

4 Efficiency Corrections

4.1 Muon Reconstruction Efficiencies

Muon reconstruction efficiencies at high p_{T} have been thoroughly studied using a data-driven tag-and-probe technique in LHCb for several analyses with Z and W bosons [29]. The subsequent Z-jet cross section analyses have studied the effect of these efficiencies on the jet kinematics and found no variation due to the presence of a jet on the away-side of a Z or W boson [21, 22]. Thus, the same efficiencies are used for this analysis and applied on a per-event basis as a function of muon pseudorapidity. The efficiency to reconstruct the dimuon pair from the Z boson is

$$\epsilon_{\mu} = \epsilon_{track}^{\mu^+} \cdot \epsilon_{ID}^{\mu^+} \cdot \epsilon_{track}^{\mu^-} \cdot \epsilon_{ID}^{\mu^-} \cdot \epsilon_{trig}^{\mu^+\mu^-}, \tag{8}$$

where ϵ_{track} is the tracking efficiency, ϵ_{ID} is the efficiency of identifying the tracks as coming from muons, and ϵ_{trig} is the total dimuon trigger efficiency. Since there are two muons present, and only one of the muons is required to TOS the trigger line, the trigger efficiency can be written as in Ref. [21] as

$$\epsilon_{trig}^{\mu^+\mu^-} = \epsilon_{trig}^{\mu^+} + \epsilon_{trig}^{\mu^-} - \epsilon_{trig}^{\mu^+} \cdot \epsilon_{trig}^{\mu^-}, \tag{9}$$

where each ϵ_{trig} is the individual trigger efficiency of the muon or anti-muon. This efficiency is calculated on a per-event basis and used to weight the total number of Z-jet pairs for the normalization of the charged hadron-in-jet distributions.

4.2 Jet Efficiencies

Jet efficiencies associated with the jet reconstruction and bin migration due to the large jet p_{T} resolution must also be determined as introduced in Eq. 7. These are calculated with the simulation samples documented in Appendix A and labeled as ϵ_{jet} and U_{jet}, respectively. The same selection cuts were applied in the simulation as in data; in particular, this includes the $\Delta\phi$ cut. Note that no dependence of these efficiencies on $\Delta\phi$ was found in the range studied here [21]. The efficiencies are defined as, similarly to Ref. [21],

$$\epsilon_{jet} \equiv \frac{N(p_T^{true} > \alpha \ \&\& \ p_T^{reco} > \beta \ \&\& \ \gamma < \eta^{true} < \delta)}{N(p_T^{true} > \alpha \ \&\& \ \gamma < \eta^{true} < \delta)} \tag{10}$$

and

$$U \equiv \frac{N(p_T^{reco} > \alpha \ \&\& \ p_T^{true} > \beta \ \&\& \ \gamma < \eta^{reco} < \delta)}{N(p_T^{true} > \alpha \ \&\& \ p_T^{reco} > \beta \ \&\& \ \gamma < \eta^{true} < \delta)}. \tag{11}$$

These definitions determine the efficiencies in a certain jet p_{T} bin, and can then be applied bin-by-bin to the data based on the measured jet p_{T}. Here, α, β, γ, and δ are general user defined cuts applied; for this analysis, the values correspond to the selection criteria discussed in Section 3 with $\alpha = 20$ GeV, $\beta = 15$ GeV, $\gamma = 2.5$, and $\delta = 4$. Reference [21] showed that applying these efficiencies, rather than purely unfolding the jet spectrum, results in the same final jet p_{T} spectrum to within 2%. It will be shown here that, with the addition of the single PV requirement, the difference between the two methods is even smaller. The method which applies the corrections bin-by-bin on a per-jet level is desired since the corrections actually have a physical meaning, rather than simply multiplying the

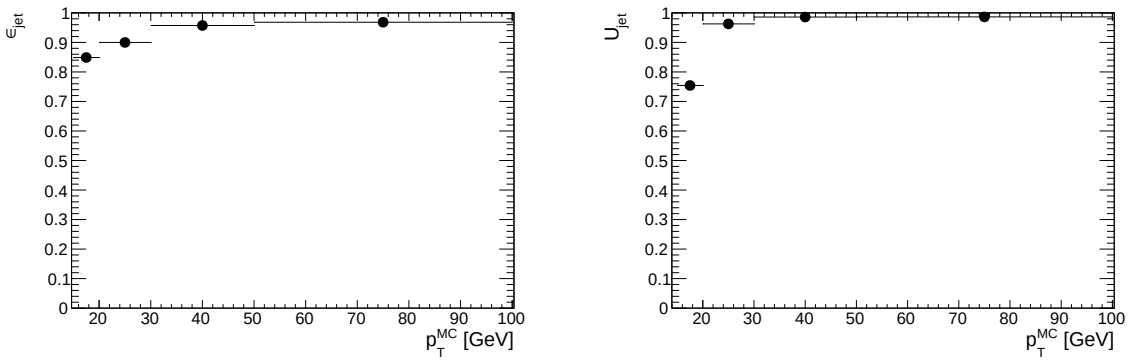

Figure 14: The jet efficiency (left) and unfold factor (right) are shown. The largest effect is seen at small p_{T}, where the jet identification requirements have a larger impact and bin migrations are larger.

number of reconstructed events by a single number which maps the total reconstructed events to the total MC events as determined from simulation.

Figure 14 (left) shows the jet efficiency ϵ_{jet} as a function of jet p_{T} in simulated Z-jet events. The efficiency is taken from simulation as discussed above and shows the expected rising feature with p_{T}. The MC jets are matched to the reconstructed jets via their ΔR separation, similarly to previous analyses [21, 22]. This is because the jet identification requirements have a larger impact at smaller p_{T}, as discussed in Refs. [21, 26]. Figure 14 (right) shows the U_{jet} unfold factor. Again a similar shape is seen due to the larger effect of bin migration at smaller jet p_{T} where the jet p_{T} resolution is larger. This is also where the bin size is smallest relative to the jet resolution, thus a higher fraction of jets may migrate into a different bin.

The bin-by-bin method was tested with the MC truth and MC reconstructed jet sample and compared to a full unfold. The unfolding was performed with the RooUnfold package [30]. An iterative Bayes method [31] is used to compare to the bin-by-bin corrections determined above. To perform the unfold, a response matrix, shown in Fig. 15, is determined from simulation and used for the unfolding procedure. To test the methods, the unfolding technique and the bin-by-bin correction are applied to the reconstructed MC jet p_{T} spectrum. Trivially, the unfolded reconstructed MC jet p_{T} spectrum matches the truth MC jet p_{T} spectrum exactly, since each was used to determine the response matrix and thus they are 100% correlated. However, Figs. 16 and 17 shows that the jet p_{T} spectrum as determined from the unfold (and thus, true MC jet p_{T} spectrum) and the bin-by-bin correction matches within 1% in the jet p_{T} range observed in this analysis. This is consistent with the findings from Ref. [21], and is actually even better due to the better jet resolution in events with only one PV. To ensure that the result was not simply due to correlations between determining the efficiencies from the same MC as they are applied to, the jet efficiencies were also determined with only the magnet down MC and then applied to the magnet up MC. Thus, they are applied to an independent data set; within larger uncertainties, the same result as Fig. 17 is found.

To determine the true p_{T} spectrum from data, the efficiency corrections determined above simply need to be applied to the reconstructed jets from data. For comparison, the response matrix can also be applied to the reconstructed jet p_{T} distribution; the results of applying each method to the jets measured in the full 2012 data are shown in

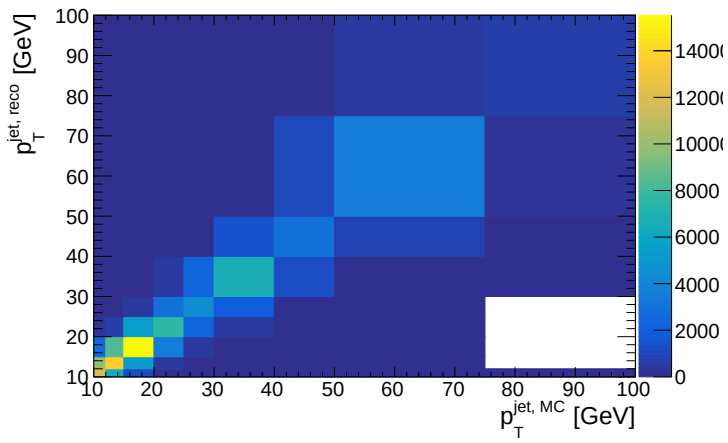

Figure 15: The jet p_T response matrix as determined from simulation. Bin migrations are evident from the response matrix. The z axis shows an arbitrary unit of counts.

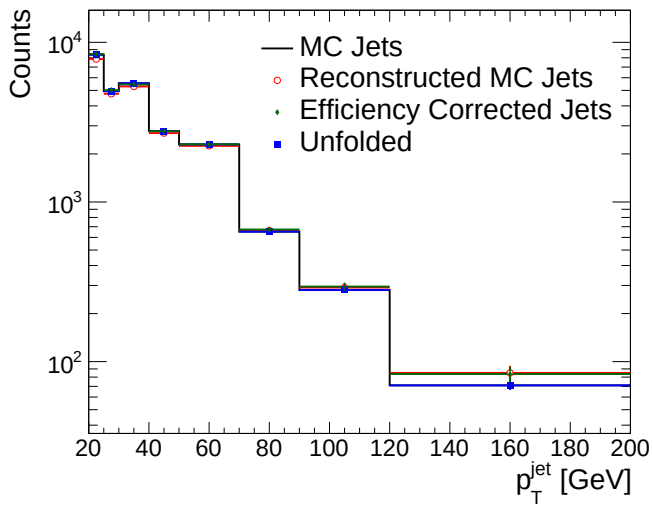

Figure 16: The jet p_T spectra for MC jets and reconstructed MC jets are shown. The unfolded reconstructed distribution is shown in addition to the bin-by-bin efficiency corrected distribution, which match within 1% as their ratio shows in Fig. 17. Note that the bins are not corrected for their varying bin widths, which is why the jet p_T spectrum looks like it is not smoothly falling. In data, jets are reconstructed up to $p_T = 100$ GeV after which there are limited statistics. Although there are differences in the reconstructed to truth MC jet spectra, the log y axis hides this difference.

Fig. 18 and both the number of jets and the uncertainties on that number are nearly identical as would be expected from the above MC study. As a reminder, the true jet p_T spectrum gives the normalization factors for the charged hadron multiplicities per-jet as discussed in Section 2. Note that Fig. 18 only shows the corrected jet counts, before the muon efficiencies are applied. This is simply because the muon efficiencies are the same regardless what method is used to determine the true jet p_T spectrum, and they amount to an approximately $\sim 20\%$ increase in the yields.

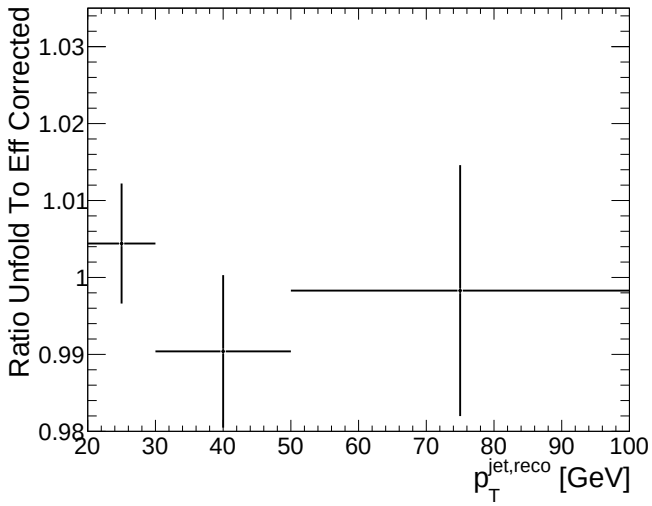

Figure 17: The ratio of the efficiency corrected to the unfolded jet p_{T} distribution is shown. The agreement between the two methods is within 1% in each jet p_{T} bin analyzed in this analysis.

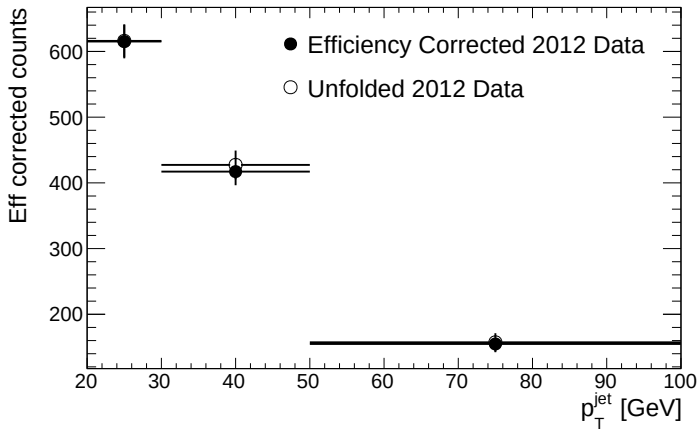

Figure 18: The efficiency and unfolding corrected counts for the true jet p_{T} spectrum is shown as determined in data. The unfolding and bin-by-bin correction techniques are consistent, as should be expected from the MC study described above.

4.3 Track Efficiencies

The charged hadron-in-jet efficiencies are also determined with the simulation. These efficiencies determine the efficiency with which tracks are reconstructed in association with and within a jet of radius $R = 0.5$. Thus, it is necessary to use simulation since the track efficiencies include the condition of being within a jet; therefore, it is necessary to have access to the truth jet information. Since the normalizing jet reconstruction efficiencies described in Section 4.2 correct the reconstructed jet spectrum to the true jet spectrum, the tracks must also be corrected back to the true charged hadron level. Therefore, the efficiencies are determined with all truth charged hadrons in truth jets, regardless of whether or not the jets were reconstructed. This way the hadron-in-jet level quantities are being corrected to the same truth level as the normalizing Z-jet spectra. Figure 19 shows a schematic diagram of the four possible options for which the track efficiencies account for when reconstructing tracks within jets. Two track efficiencies can be defined

which account for these four possibilities. The $\epsilon_{track}^{in\ jet}$ efficiency accounts for tracks that are not reconstructed within the jet cone, shown by the thick blue, dotted purple, and thin green arrows. The $\epsilon_{track}^{gained}$ efficiency accounts for truth tracks outside of the truth jet which are misreconstructed within the reconstructed jet, shown by the hatched red arrow.

Figure 19: A schematic cartoon showing the various outcomes of tracks due to track misreconstruction or efficiency. Tracks could either be reconstructed correctly (purple dotted), not reconstructed at all (thick blue), or misreconstructed outside of the jet cone (thin green); these tracks are accounted for with $\epsilon_{track}^{in\ jet}$. Truth tracks outside the truth jet cone which are misreconstructed within the jet are shown by the hatched red arrow, which are accounted for with $\epsilon_{track}^{gained}$.

The efficiencies are determined two dimensionally in p, η space similarly to Ref. [23]. The $\epsilon_{track}^{in\ jet}$ efficiency accounts for tracks inside the truth jet which are not reconstructed within the reconstructed jet. Truth charged hadrons within the truth jet can either be misreconstructed outside of the reconstructed jet or not reconstructed at all, where the truth charged hadrons have already been identified as jet constituents by the MC particle flow algorithm. This efficiency then accounts for the lost tracks, and is defined as

$$\epsilon_{track}^{in\ jet} = \frac{N_{reco}(\Delta R(reco\ track, reco\ jet) < R)}{N_{truth}(\Delta R(truth\ jet, truth\ track) < R)}, \tag{12}$$

where R is the jet radius and ΔR is the distance in η, ϕ space between the denoted objects. The second efficiency considers the number of tracks that are gained within the reconstructed jet due to track misreconstruction; for example, a truth track that lies outside of the truth jet may get reconstructed within the reconstructed jet and thus count as a reconstructed jet constituent. This efficiency then accounts for the gained tracks and can be defined as

$$\epsilon_{track}^{gained} = \frac{N_{reco}(\Delta R(reco\ track, reco\ jet) < R\ \&\&\ \Delta R(truth\ track, truth\ jet) > 0.5)}{N_{reco}(\Delta R(reco\ track, reco\ jet) < R)}, \tag{13}$$

where again R is the jet radius and ΔR is the distance in η, ϕ space between the denoted objects. In this efficiency, the track matching condition in the numerator is determined via the standard `DaVinciSmartAssociator` LHCb reconstructed-truth track matching algorithm; this requires that at least 70% of the LHCbIDs match to the truth track. This value then quantifies the percentage of reconstructed tracks in a given track p, η bin that originate from truth tracks outside the truth jet, and thus should be removed from the

overall number of tracks. Because of the way this efficiency is defined, the number of good tracks is then $1 - \epsilon_{track}^{gained}$, making the total track efficiency

$$\epsilon_{track}^{total} = \frac{\epsilon_{track}^{in\ jet}}{1 - \epsilon_{track}^{gained}} \tag{14}$$

such that $N_{true}^{track} = N_{reco}^{track} / \epsilon_{track}^{total}$.

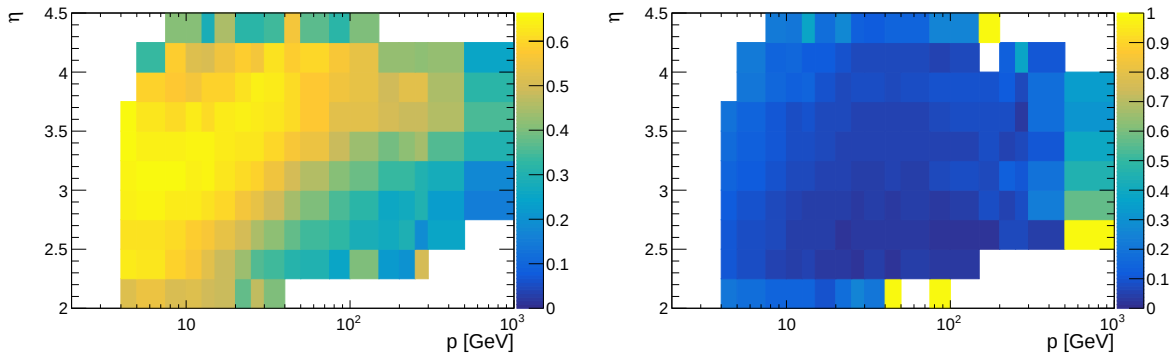

Figure 20: The $\epsilon_{track}^{in\ jet}$ (left) and $\epsilon_{track}^{gained}$ (right) efficiencies are shown as a function of p and η of the track.

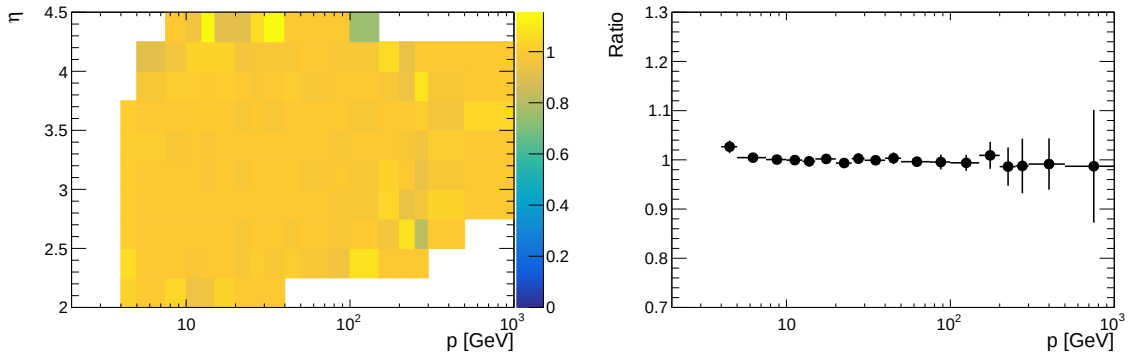

Figure 21: The ratio of efficiency corrected reconstructed MC tracks-in-jets to truth MC tracks-in-jets is shown as a function of η and p (left). The ratio is consistent with unity across all p and η. The bins which appear to be far from unity have low occupancy and thus large uncertainties, and are near the edge of the fiducial area of this measurement. The projection onto the x axis, or momentum axis, is shown which demonstrates this consistency (right).

The efficiencies are determined in simulation with the same selection criteria for the Z and jet used to determine the jet efficiencies as described in Section 3. The $\epsilon_{track}^{in\ jet}$ is shown in the left panel of Fig. 20 as a function of the momentum and pseudorapidity of the track, and the $\epsilon_{track}^{gained}$ is shown in the right panel of Fig. 20. The $\epsilon_{track}^{in\ jet}$ efficiency is large up to several hundred GeV in p, after which the efficiency begins to fall off due to tracks having a smaller bending radius and thus being more difficult to reconstruct. In particular, this decrease in the efficiency with momentum is due to a cut in the Particle Flow algorithm used by LHCb to reconstruct jets which requires that the error on the bending radius of the track is small. The $\epsilon_{track}^{gained}$ efficiency is small over the range of this measurement (on average approximately $\sim$5%), however it is clear that it is nonzero and

needs to be accounted for. The number of tracks gained in the reconstructed jet gets larger with increasing p, where again tracks are more likely to be misreconstructed due to their small bending radius.

To ensure that the drop in efficiency of track reconstruction within jets from the bending radius cut is well modeled in simulation, a short study was performed with inclusive $Z \to \mu\mu$ decays. The `sqrt(PERR2)/P` distributions were studied for both muons in simulation and data to ensure that the uncertainty on the bending radius, which depends on the uncertainty of the momentum, is well modeled in both simulation and data. Figure 22 shows the results for positive and negative muons in two momentum bins at high momentum where the track-in-jet efficiency drops off.

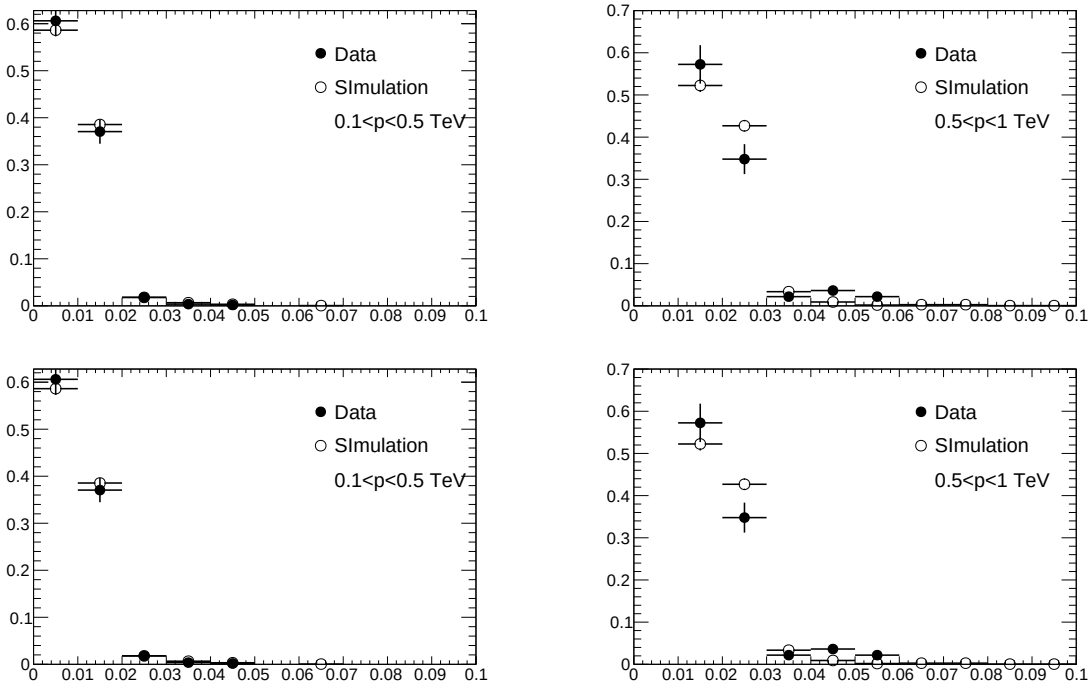

Figure 22: The data and simulation matching for decay μ^- (top) and μ^+ (bottom) `sqrt(PERR2)/P` in inclusive Z decays is shown in two high momentum bins. The matching is good and displays that the bending radius uncertainty modeling in simulation is good in the determination for high momentum tracks.

To test the efficiencies, they are used to correct the reconstructed MC charged hadrons-in-jets and compared to the truth charged hadron-in-jet distributions. The ratio of these distributions is shown in Fig. 21, and is consistent with unity across all momentum and pseudorapidity. A 1D projection onto the momentum axis is also shown in Fig. 21 (right), which more clearly shows the consistency within the uncertainties. The ratio of efficiency corrected reconstructed MC tracks to truth MC tracks as a function of p_{T} can also test that both the momentum and pseudorapidity dependence of the efficiencies is correct. This is shown in Fig. 23, which also shows that the ratio is consistently unity across all p_{T} analyzed. Note that there is some self-correlation here, since the efficiencies are determined with the same set of MC data that is being tested. To avoid this auto-correlation, a separate test was performed with the efficiencies determined in the magnet down MC and applied to the magnet up MC. Since these are two independent MC data sets, this

is a better test of the method. Figure 24 shows that the same conclusions hold when determining the efficiencies with the magnet down MC and applying them to the magnet up MC, within the larger uncertainties.

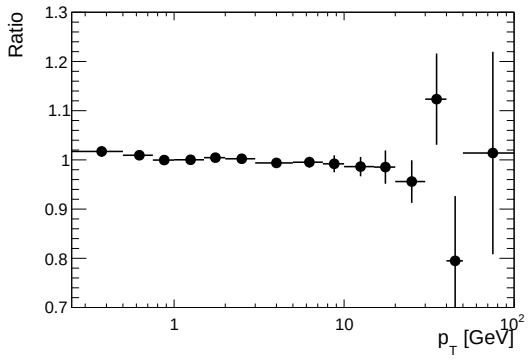

Figure 23: The ratio of efficiency corrected reconstructed MC tracks-in-jets to truth MC tracks-in-jets is shown as a function of p_{T}, which is consistent with unity across all p_{T} in this analysis.

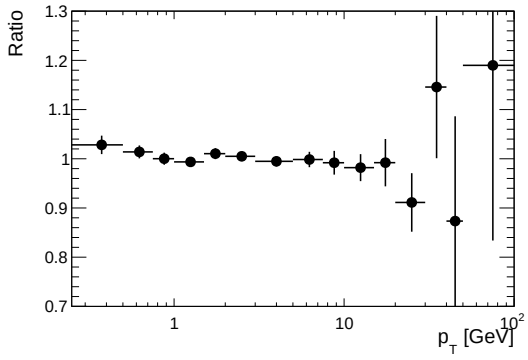

Figure 24: The ratio of efficiency corrected reconstructed MC tracks-in-jets to truth MC tracks-in-jets is shown as a function of p_{T}. Here, the efficiencies are determined with only the magnet down MC and then applied to only the magnet up MC. This allows for the efficiencies to be determined and applied in an independent way, providing an additional cross check of the method.

Figure 25 shows the total track efficiency as a function of the three kinematic variables analyzed for the 2012 data set, integrated over jet p_{T}. Note that no significant variation of the track efficiencies was observed with the jet p_{T} within the statistical precision of the MC sample. The total track efficiency is consistently smaller at small r, large j_T, and large z, which correspond to high momentum particles where the $\epsilon_{track}^{in\ jet}$ is smaller and the $\epsilon_{track}^{gained}$ is larger.

4.4 SPD Hit Multiplicity

In previous Z-jet cross section analyses, a correction has been applied due to the SPD hit requirement of the trigger line used in these analyses [21, 22]. The `L0Muon` trigger line has a requirement that the number of hits in the SPD must be less than 600; this

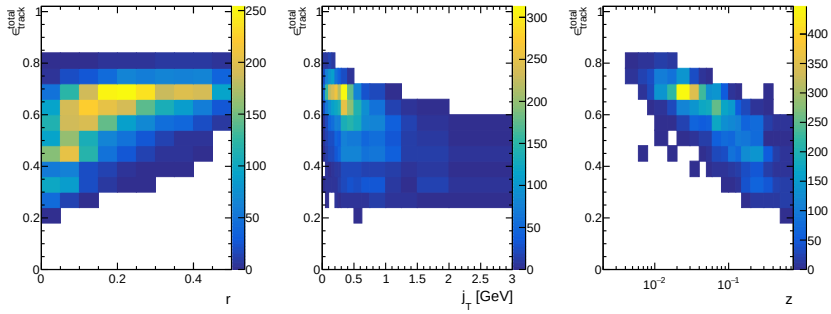

Figure 25: The total track efficiency is shown on the y axis as a function of three track-in-jet variables analyzed. The z axis is the number of charged hadrons in data that fall in a particular bin of total track efficiency vs. z (or vs. j_T or r). The figures show the efficiencies integrated over jet p_{T}.

condition is placed to keep the event size small and thus the processing times for these events reasonable. This introduces a bias into the Z-jet cross section analyses, since there are Z-jet events which in principle might have an SPD count greater than 600. Thus, the analyses apply a correction for this bias using the `L0DiMuon` trigger line which does not have the SPD hit requirement. The `L0DiMuon` trigger line has been shown to reproduce the `L0Muon` trigger line SPD hit distribution, and thus it can be used to estimate the number of events which are unobserved due to the trigger requirement. Note that applying this correction is better than using the `L0DiMuon` trigger line due to the higher efficiency of the single muon line, as explained in Refs. [21, 22].

Since this analysis uses the single muon trigger line, a check was performed to ensure that events were not being neglected with SPD hit multiplicities greater than 600. This study is similar to the previous studies in that the SPD hit multiplicities were studied for both the `L0Muon` and `L0DiMuon` trigger lines. The SPD hit multiplicities for both trigger lines are shown in Fig. 26, where the histograms are filled if the particular `L0` trigger is TOS. The distributions are consistent, as should be expected based on the results of Refs. [21, 22]. However, Fig. 26 shows that both trigger lines do not have events where the SPD hit multiplicity is greater than 600. This is because of the condition that only 1 PV must exist, which the previous Z-jet analyses did not specify; therefore, the number of events with a large number of SPD hits is significantly reduced. This shows that there is not a bias introduced in the number of charged hadrons in a jet because of the trigger line chosen, and thus we do not apply any correction for the SPD hit multiplicity like the previous Z-jet cross section analyses.

4.5 Efficiency Corrected Distributions

The track efficiencies are then applied to the hadron-in-jet distributions in the 2012 data to determine the efficiency corrected distributions which will be unfolded. These distributions are shown in Fig. 27. The distributions are all scaled by their respective bin widths. These distributions show general perturbative QCD expectations as the jet p_{T} increases; for example, the j_T spectrum becomes harder as the jet p_{T} increases which follows expectations from TMD Q^2 evolution predictions [32, 33]. The distributions also follow general expectations from constituent in jet distributions; for example, the z range

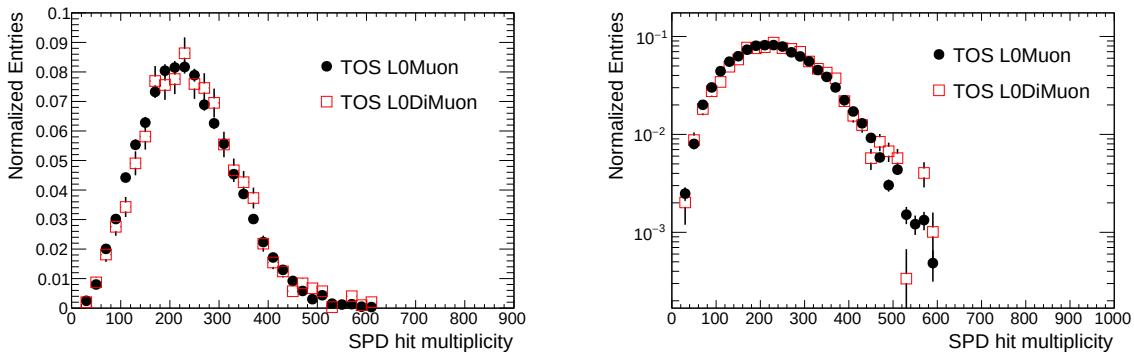

Figure 26: The number of SPD hits is shown for two trigger lines, the `L0Muon` and `L0DiMuon` TOS lines on a linear scale (left) and a log scale (right). The distributions are the same and show that there are not a significant number of events with SPD hits greater than 600 in the single muon trigger line, which is due to the condition of 1 PV in the event.

probed increases and the jet becomes more collimated as the jet p_{T} increases.

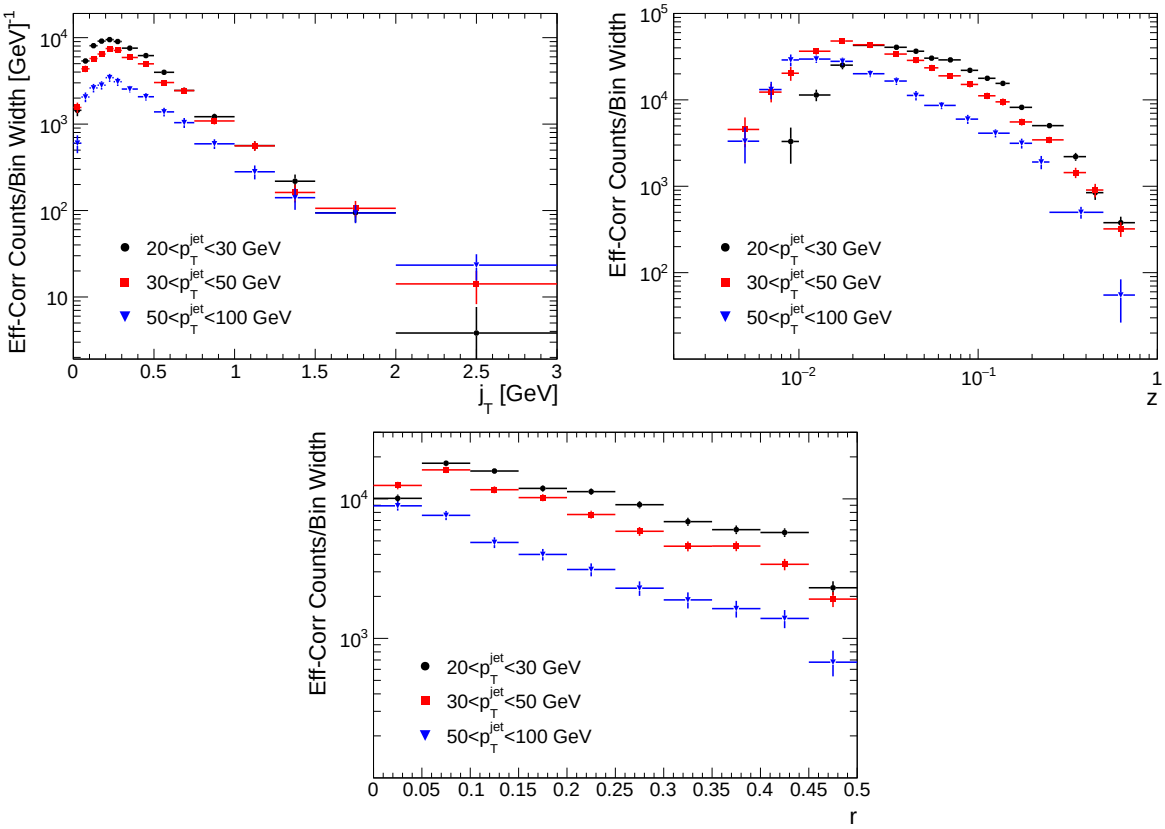

Figure 27: The efficiency corrected charged hadron-in-jet distributions are shown, scaled by their respective bin widths. The longitudinal momentum fraction z, transverse momentum j_T, and radial profile r are shown and display jet substructure features.

As mentioned in Section 1, several other observables are also measured for comparisons to the three "main" observables of this analysis. Since one goal is to measure the charged hadron distributions in as many different ways as possible, the observables p^{track}, p_T^{track}

and θ^{track} provide potentially different approaches to looking at the hadronic structure of jets. These distributions, track-efficiency corrected, can be seen in Fig. 28 and display similar qualities to the z, j_T, and r distributions. Figure 28 also shows the $\theta^{track}_{weighted}$ distributions, where the definition of the observable is similar to θ^{track} except that each hadron is weighted by their p_{T}. This observable may give insight into the angular ordering or radiation patterns of the jet substructure at the hadron level, and is similar to other p_{T} weighted jet substructure observables like the jet pull vector [34] or the jet girth [35]. However, it is important to emphasize that observables like the jet pull or jet girth are on a jet-by-jet basis, whereas the observables shown here are on a hadron-by-hadron basis and thus provide different but complementary information to other jet substructure observables.

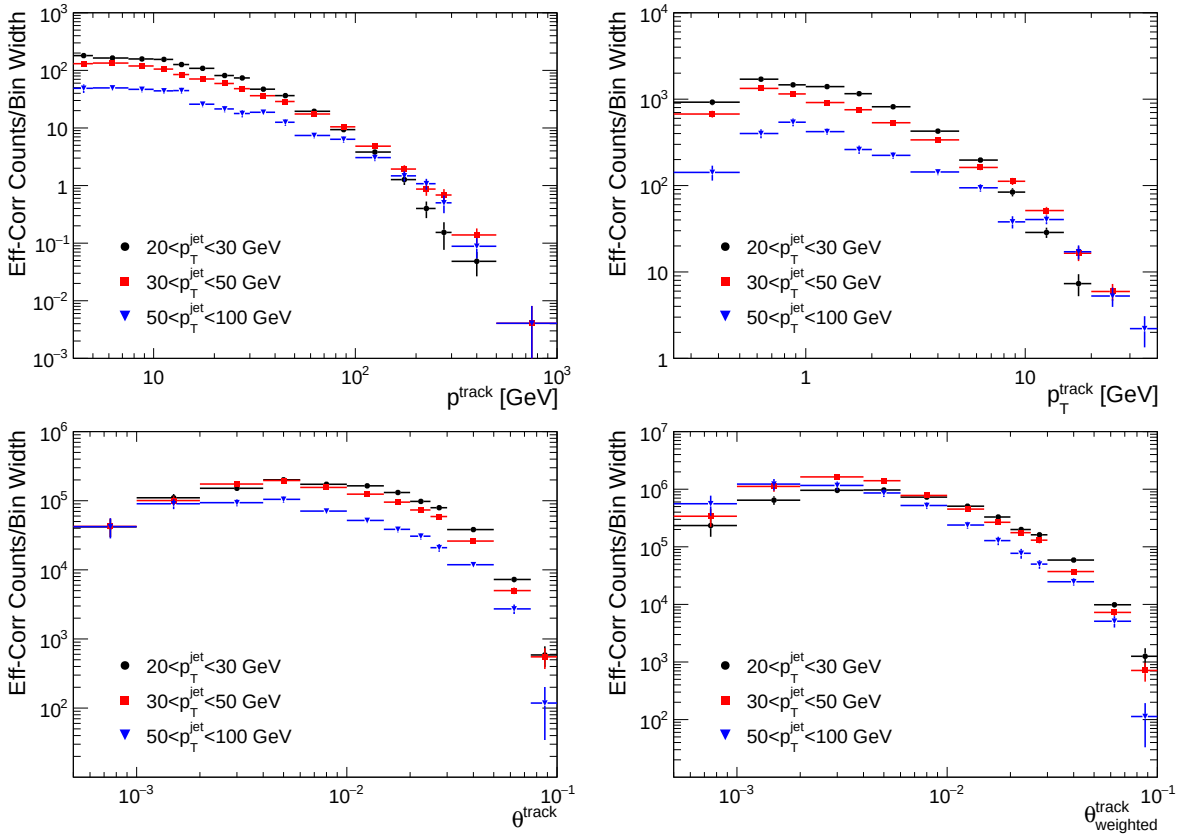

Figure 28: The efficiency corrected charged hadron-in-jet distributions are shown, scaled by their respective bin widths. Here the track momentum p, p_{T}, θ^{track}, and $\theta^{track}_{weighted}$ variables are shown. They display similar qualitative features to z, j_T, and r as might be expected since they are related via various kinematic and/or trigonometric relations.

5 Unfolding

A 2D unfolding must be performed on the efficiency corrected charged hadron-in-jet distributions to account for the large jet p_{T}, η, and ϕ resolution that can cause bin migrations in the jet level quantities and consequently the values of the charged hadron with respect to jet quantities like z. Note that any migration is almost entirely due to the

jet reconstruction, since, for example, the track p resolution at LHCb is generally less than 1%. The unfolding procedure then accounts only for the bin migrations, since effects from lost and/or misreconstructed tracks are already considered from the efficiencies. Response matrices need to be constructed for each of the kinematic observables measured in this analysis.

The MC sample is used to construct the response matrices and determine the effect from the smearing. Reconstructed jets are matched to truth jets via their ΔR separation, similarly to what was done for determining the jet efficiencies. Reconstructed tracks are matched to truth tracks with the standard `DaVinciSmartAssociator`, again similarly to what was done for determining the track efficiencies. Only tracks which are appropriately matched are analyzed; that is, reconstructed tracks are considered only if they match a truth track that is within the truth jet. This is to ensure that we only consider the effects from smearing, and to not double count effects already considered from the track efficiencies.

Example response matrices for several kinematic variables are shown in Figs. 29, 30, and 31. The response matrix shows that a large fraction of events are properly reconstructed; note the log z scale used in the plot. However, a non negligible number are smeared out which shows the necessity for the unfolding procedure. The response in z shows expected kinematic features; for example, when the reconstructed jet p_{T} is larger than the true jet p_{T} the reconstructed z is smaller than the true z. Interestingly, the smearing in the variables that only depend on the jet angle, j_T and r, is larger than that in z. This can kinematically be understood as due to the smearing *relative* to the bin size examined. While the jet resolution in ϕ and η is actually smaller than the jet resolution in p_{T}, as shown in Figs. 7 and 8, the bin sizes in r and j_T relative to this resolution are smaller and thus there is more smearing in j_T and r than in z.

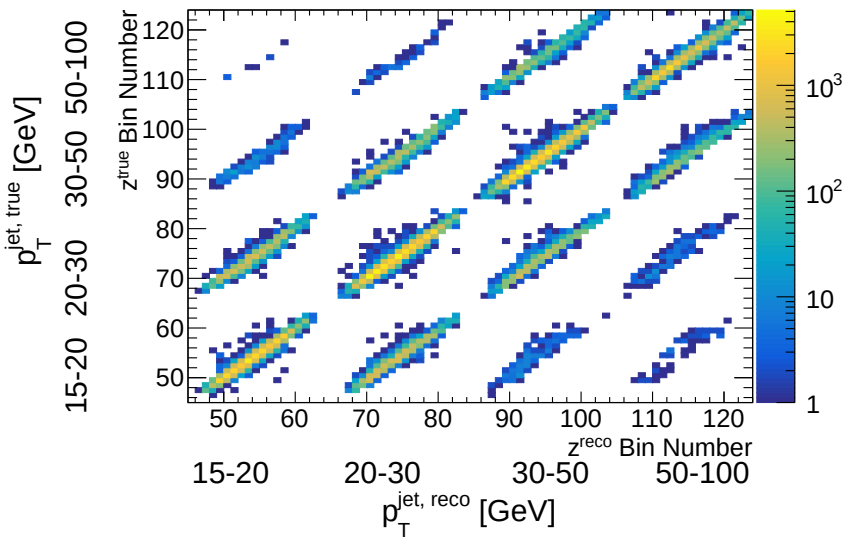

Figure 29: The 2D response matrix for the fragmentation variable z is shown as a function of the bin number. The z axis shows an arbitrary unit of counts.

To show the relative amount of smearing that occurs, Fig. 32 shows the ratio of the true distribution to the smeared distribution from the MC used to construct the response matrices. The figures show that the jet resolution causes bin migrations typically less

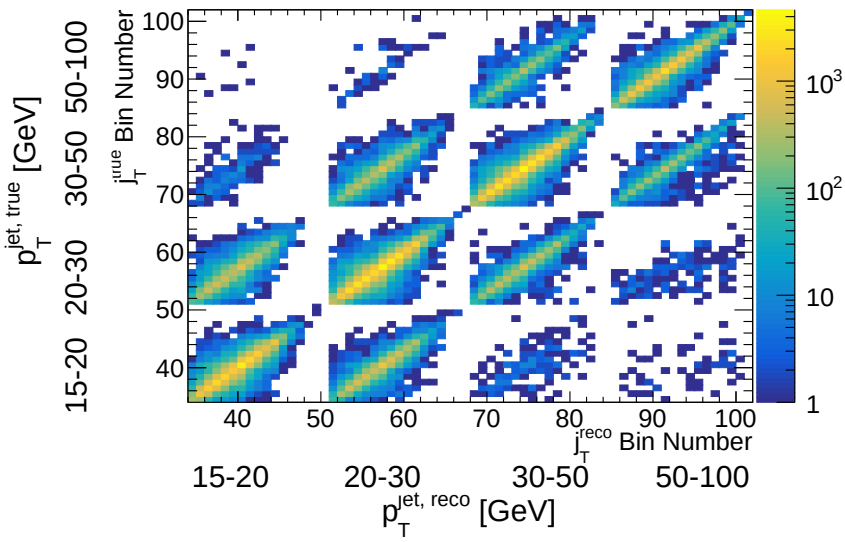

Figure 30: The 2D response matrix for the fragmentation variable j_T is shown as a function of the bin number. The z axis shows an arbitrary unit of counts.

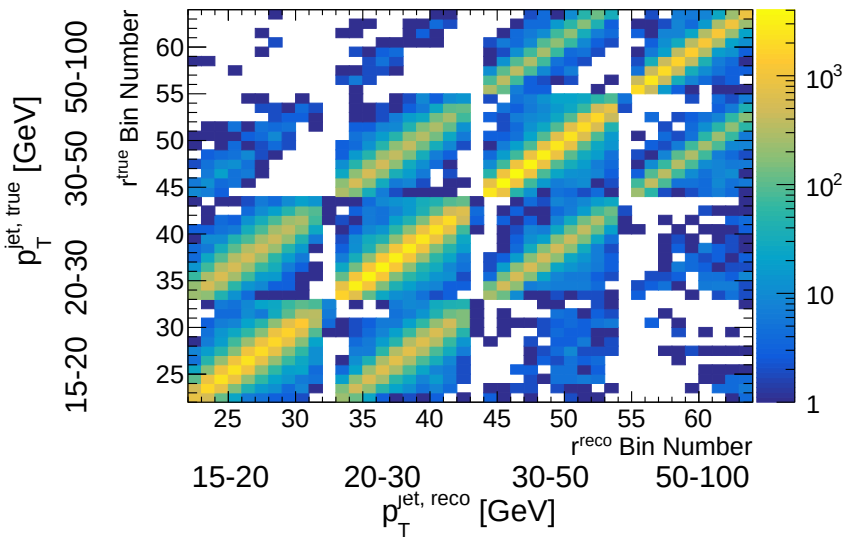

Figure 31: The 2D response matrix for the fragmentation variable r is shown as a function of the bin number. The z axis shows an arbitrary unit of counts.

than ∼5%, except at large and low z where the bin migrations can be approximately 10%.

When applied to the data, the unfolding matrices are determined with the full MC sample so as to take full advantage of the statistics available. An iterative Bayes unfolding procedure [31] is implemented with 7 iterations, where the Bayes unfolding is implemented using the RooUnfold package [30]. The raw, efficiency corrected, and unfolded counts as a function of z, j_T, and r, are shown for the three jet p_{T} bins analyzed in the following figures. These distributions are then scaled by the total number of N_{Z+jet} in a given jet p_{T} bin as described in the Introduction and shown in Fig. 36, which shows the number of hadrons per jet in Z-jet events for a given observable.

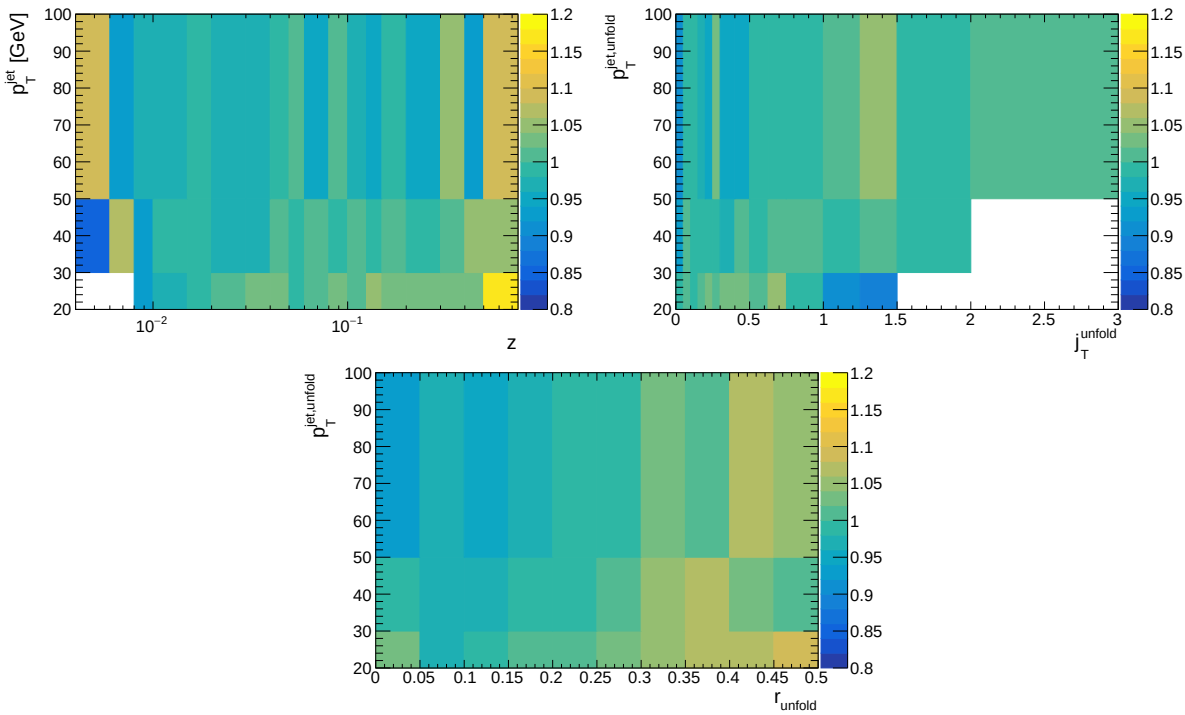

Figure 32: The ratio of the truth to smeared distributions in MC is shown for z, j_T, and r. Here the z axis corresponds to this ratio; thus, if there is no smearing, the bin value is unity. These demonstrate the approximate percentage effect that bin migration has on the various kinematic variables, which is generally less than 5%.

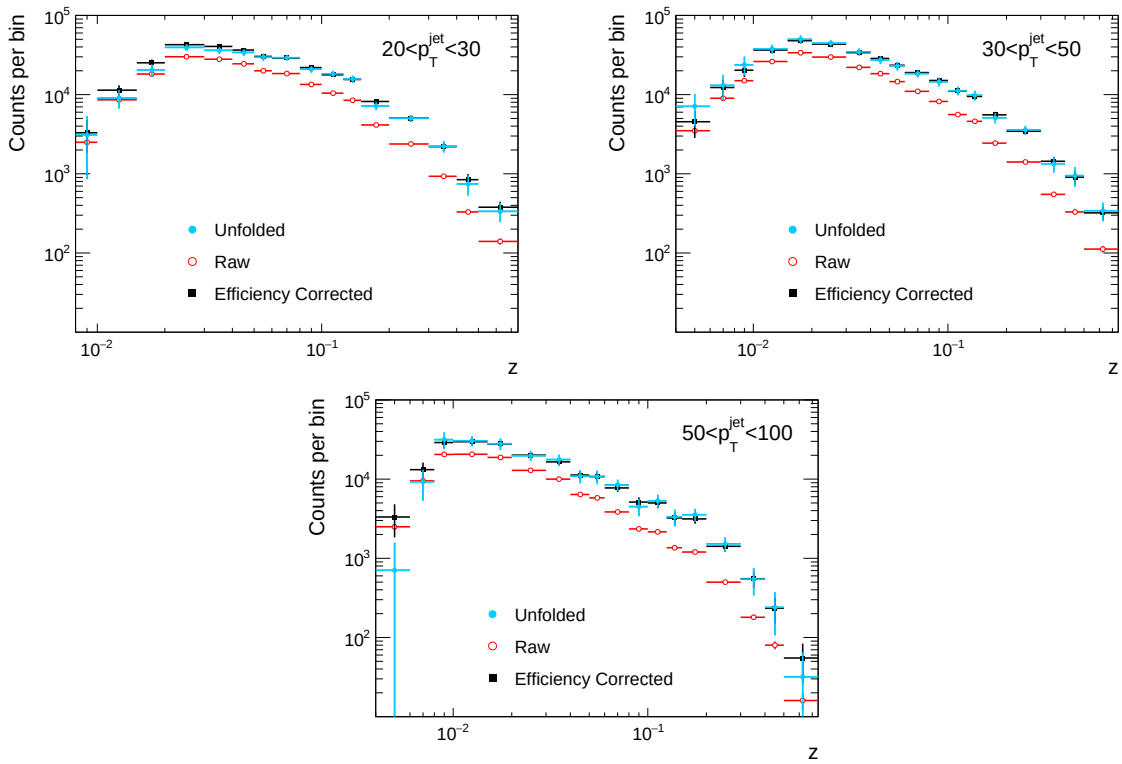

Figure 33: The raw, efficiency corrected, and unfolded counts are shown for three jet p_{T} bins as a function of z.

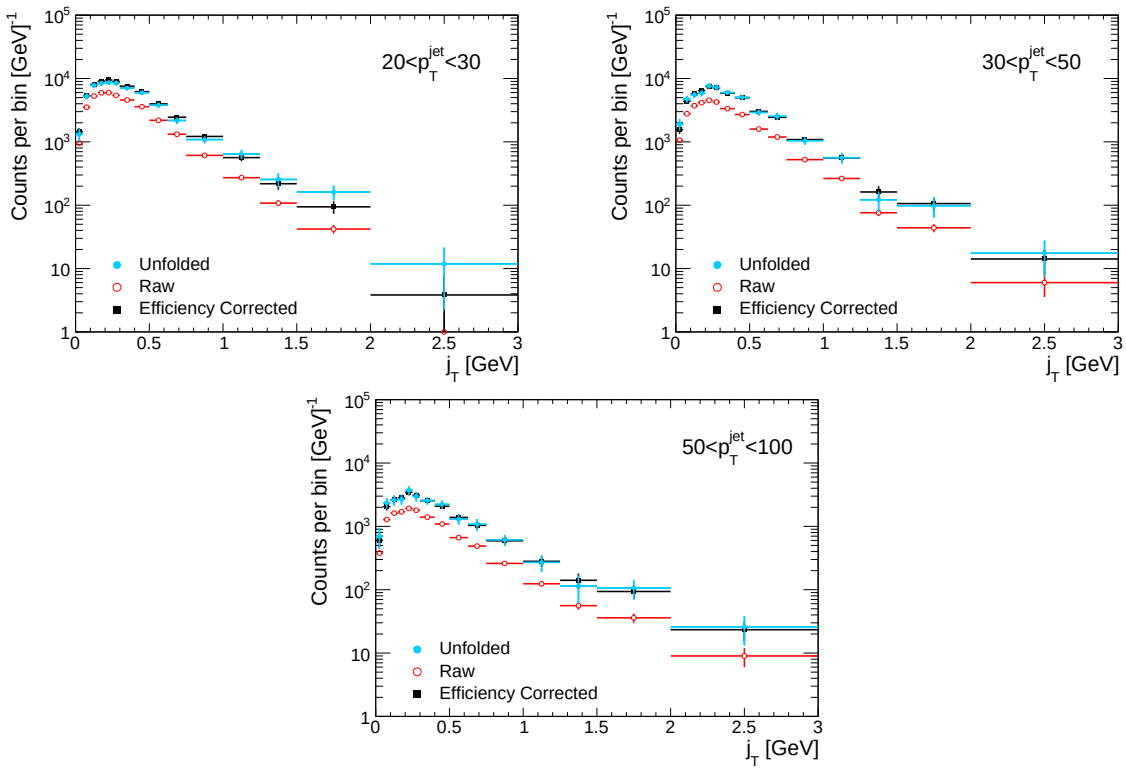

Figure 34: The raw, efficiency corrected, and unfolded counts are shown for three jet p_{T} bins as a function of j_T.

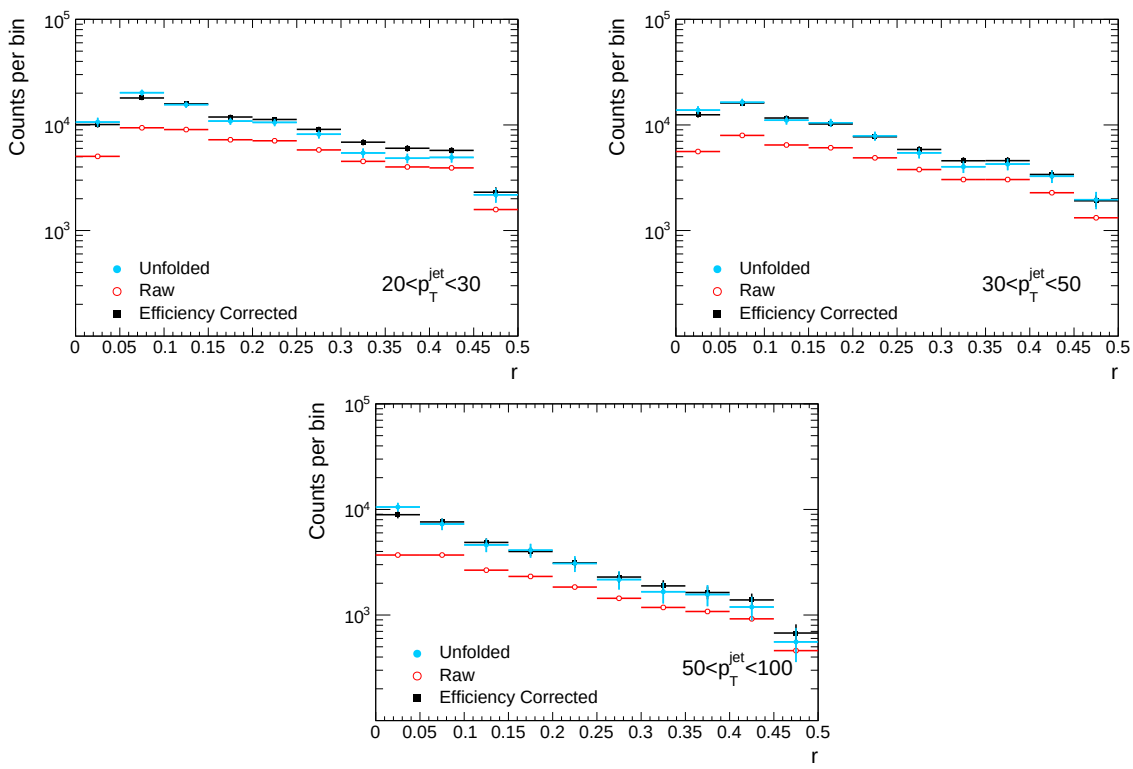

Figure 35: The raw, efficiency corrected, and unfolded counts are shown for three jet p_{T} bins as a function of r.

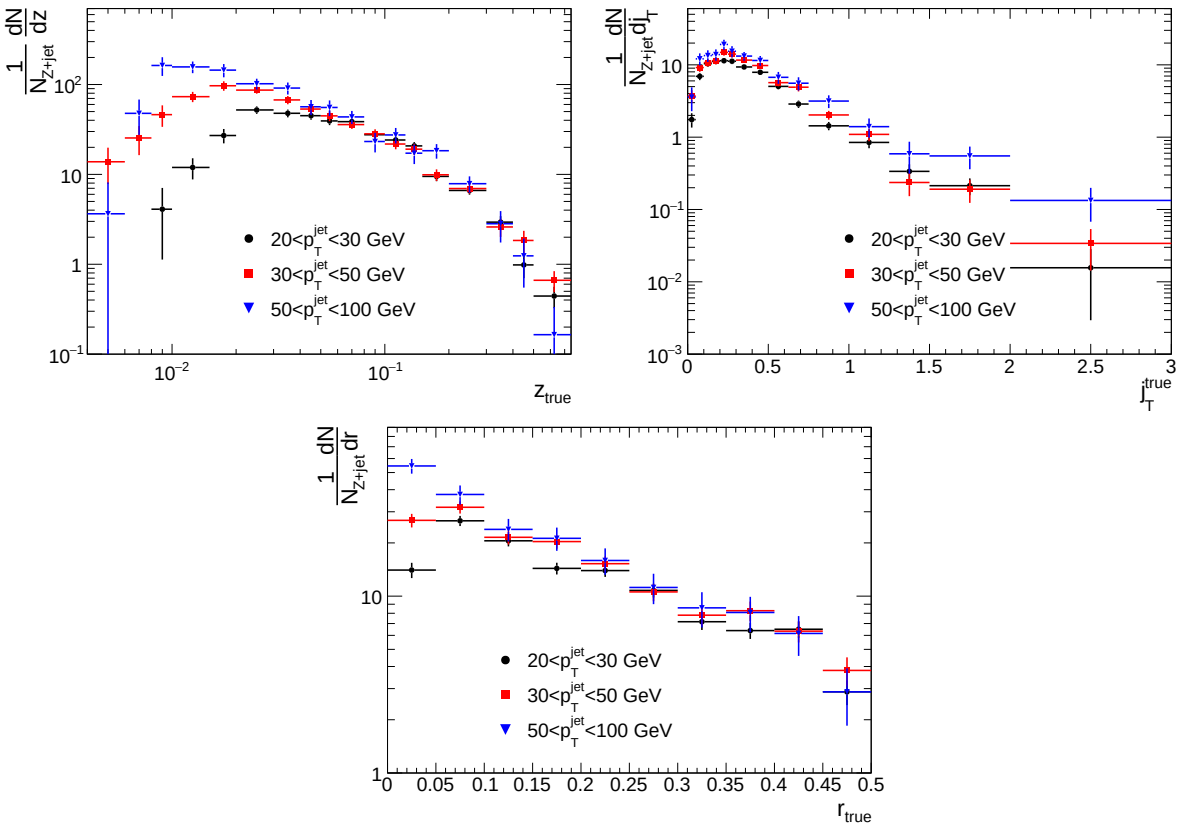

Figure 36: The final unfolded distributions scaled by the number of Z-jet pairs in a given jet p_{T} bin, shown with statistical uncertainties only.

6 Systematic Uncertainties

6.1 Overall Normalization Uncertainties

The following uncertainties only apply to the overall normalization N_{Z+jet} and are thus point-to-point correlated in each jet p_{T} bin. They amount to an overall shift of the yields by some factor, in which the factor would apply equally to all bins either up or down.

6.1.1 Muon Reconstruction Efficiencies

Uncertainties for the muon reconstruction efficiencies are assigned based on the precision with which they are determined and the method with which they are determined, as in the inclusive and Z-jet cross section analyses [21, 25]. Systematic uncertainties are determined by re-evaluating the total number of Z-jets in a given bin where the total muon efficiency is varied by one standard deviation around the central value. This changes the total normalization by 0.7%, which is added as an uncertainty and is constant as a function of jet p_{T}.

6.1.2 Jet Reconstruction Efficiencies

The jet reconstruction efficiencies for jets which pass the standard ID cuts in LHCb have been evaluated the same way for both the $\sqrt{s} = 7$ and 8 TeV analyses [21, 22]. They are estimated by tightening the jet ID cuts used to identify the jets and comparing the agreement between simulation and data. In the Z-jet cross section analysis, this uncertainty was found to be 1.9% [21]; the same procedure was replicated here and for the various jet ID variables the uncertainty varied around 1.9%, within larger uncertainties due to the PV vertex cut in this analysis. Thus, a 1.9% uncertainty is also assigned here to account for the jet reconstruction efficiency which is dominated by the jet ID cuts.

The statistical uncertainties on the determination of the efficiencies are also examined as a possible source of systematic uncertainty. When the efficiencies are changed by one standard deviation, similarly to the muon efficiencies, the change in the resulting number of Z-jets is less than 0.1% in all bins; thus, this uncertainty is neglected due to its size.

We also examine the possibility of the standard LHCb jet identification cuts biasing our charged track distributions. For example, the requirement is placed at the jet ID level that a jet can not have a particle that contains more than 80% of the jet energy. This could clearly bias the measurement at high z if there are jets that would in principle pass this criterion but are rejected at the reconstruction stage due to the jet ID cut.

Figure 37 shows the jet MPT, CPF, and MTF distributions for jets which pass the selection criteria in this analysis but are not required to pass the jet ID cuts. The MPT, CPF, and MTF cuts are defined as the the requirement that the jet contains at least one track with $p_{\mathrm{T}} > 1.2$ GeV, the charged particle fraction of the jet be at least 10%, and the requirement that no one particle can carry more than 80% of the jet energy. To estimate to what percent the distributions measured in this analysis may be biased by cutting these regions out, the distributions are integrated in their "cut region" and compared to the integral of the full distribution. For example, the CPF distribution is integrated between [0,0.1] and compared to the full integral over the entire distribution to estimate the number of jets cut out by the jet ID cuts. The number of jets removed by the cuts varies between 1-3% depending on the particular observable. Since similar

results were found with the previous study where the jet ID cuts were tightened and a 1.9% uncertainty is already applied accounting for this, this suffices to account for any bias the distributions may encounter from these jet ID cuts.

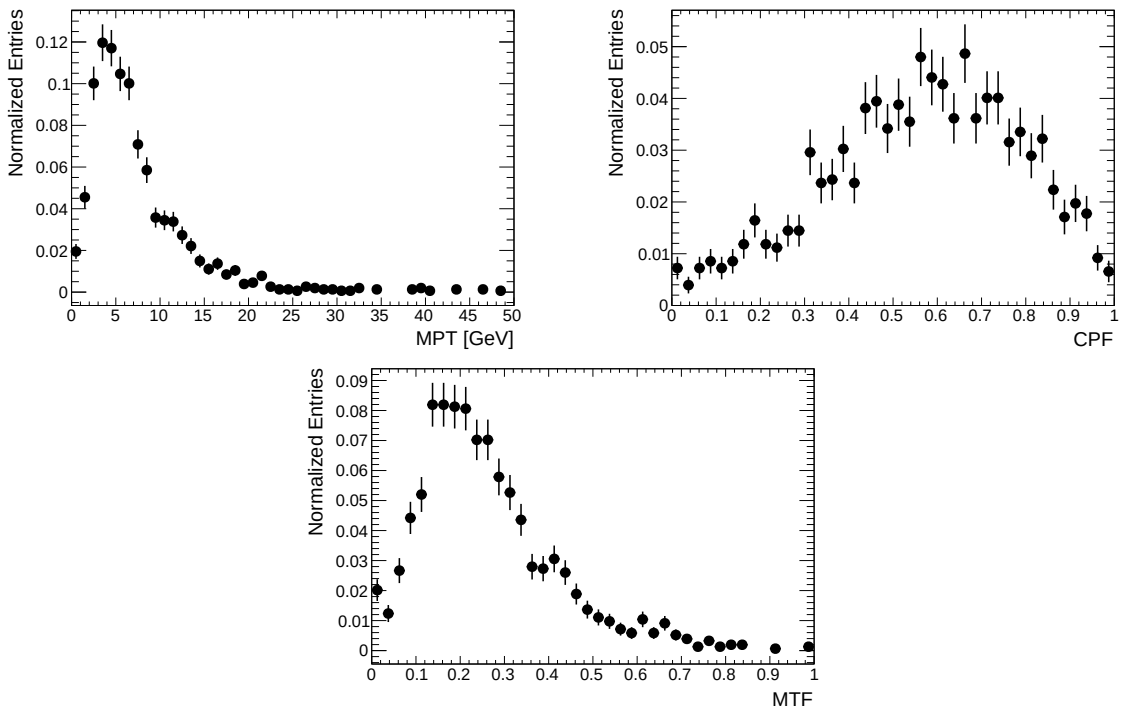

Figure 37: The jet MPT, CPF, and MTF distributions are shown for the jet selection criteria in this analysis without the requirement of the jet ID cuts. The integral of the counts in the regions that are cut out by the jet ID cuts to the total distributions ranges from 1-3%, depending on the observable.

6.1.3 Z-jet Selection Purity

The selection purity of this analysis is incredibly high, however it is not exactly unity; thus, a systematic uncertainty is assigned to account for this. The selection purity can be affected by both the purity of the Z and jet. The fake jet rate is determined in a similar fashion to Ref. [21]. Fake jets are determined in simulation by counting the number of reconstructed jets that pass the selection criterion but do not have an associated truth jet matched to them. The fake jet rate is 0.3%, which is better than the cross section analysis because of the additional requirement in this analysis of one PV. Since the selection criteria used to identify Z bosons is the same as Ref. [21], the purity of the Z can be taken to be the same as the previous analysis as $(98.6 \pm 0.3)\%$. Therefore, the impurity is $(1.4 \pm 0.3)\%$, so the total impurity for the Z-jet yields is just $\rho_Z \times \rho_{jet} = 0.986 \times (1 - 0.003) = 0.983$, where ρ is the purity for a given object. Thus the impurity is 1.7% and is assigned as a systematic uncertainty. Note that we assume these purities to be independent; it is possible that they are correlated and thus that this impurity is smaller. However, this will be a small contribution to the total systematic uncertainties, and regardless does not change the final conclusions.

6.1.4 Pile Up

Contributions from pile up were considered as a possible source of systematic uncertainty in the measurement. To estimate the contributions from pile up, we measure the charged hadron-in-jet distributions for events where 1 PV is reconstructed and separately in events where 2 PVs are reconstructed. The ratios of these distributions are consistent with unity within uncertainties, and are shown in Fig. 38. To estimate a possible deviation from unity, we fit the ratios to constant functions which yields a 2% difference. To estimate the systematic uncertainty from pollution from events which have more than 1 PV, we find the ratio of events with 1 reconstructed PV and greater than 2 truth PVs to greater than 2 truth PVs in the MC, defined as

$$Ratio = \frac{N_{events}(1\ PV_{reco}, > 2\ PV_{truth})}{N_{events}(> 2\ PV_{truth})}. \tag{15}$$

This tells us the percentage of times we reconstruct 1 PV, but actually had more than 1 PV at truth level, and thus is the pollution of this 2% difference. This ratio is 15%, and thus the pile up uncertainty is 15% of 2%, which is 0.3%. Thus, we consider it to be negligible.

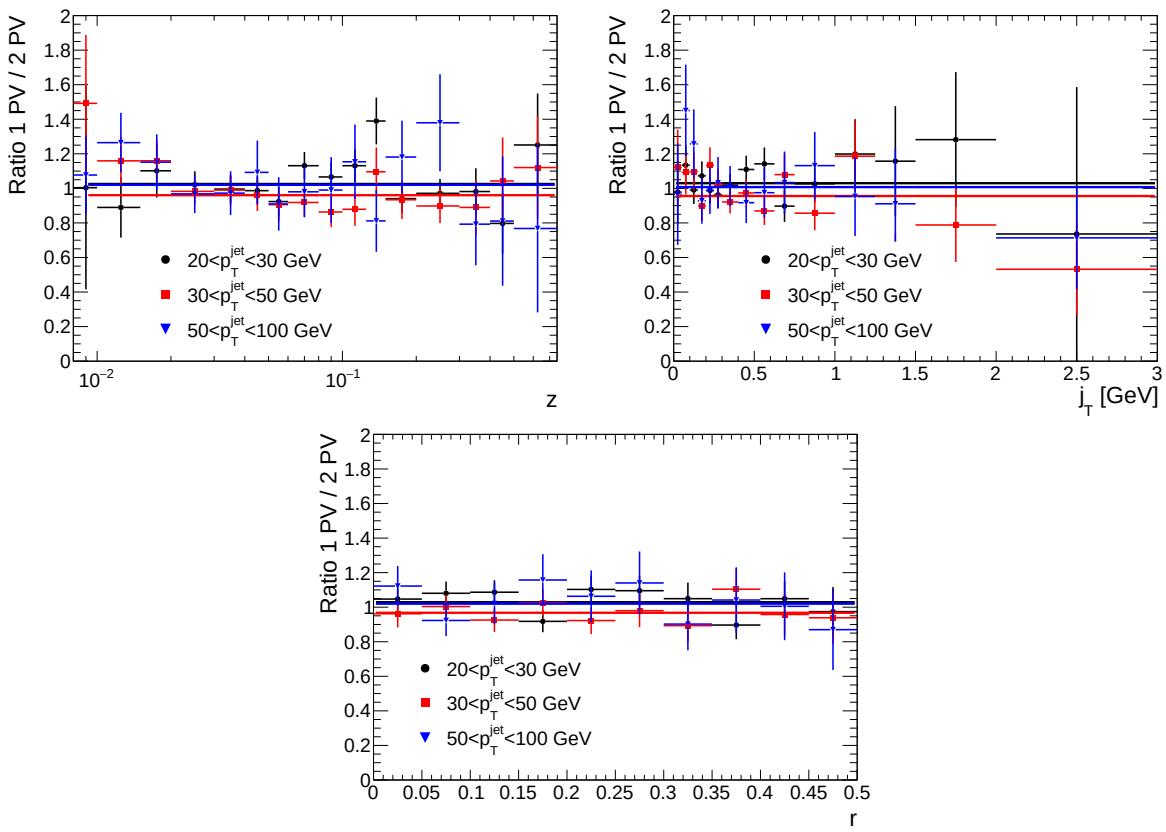

Figure 38: The ratio of the distributions with 1 PV required and with 2 PVs required. The ratios are consistent with unity, albeit within large uncertainties. Constant fits are shown in the same color as the various jet p_T bin data points, which are all consistent with unity within the uncertainty of the fit.

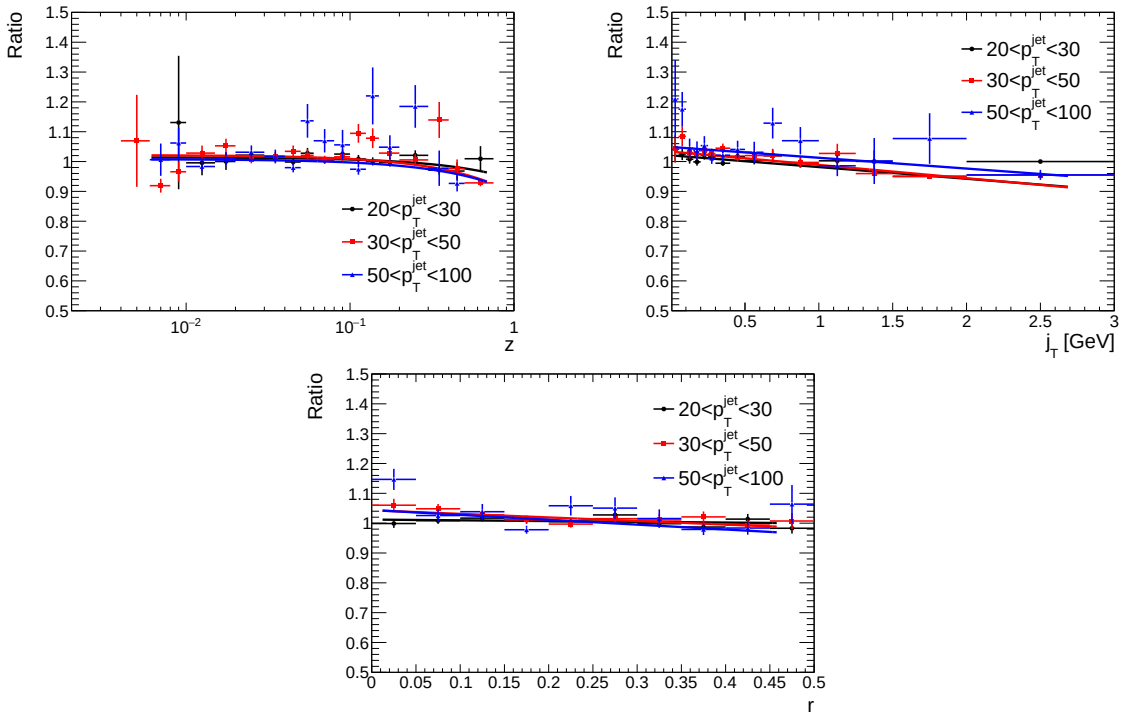

Figure 39: The ratio of the distributions with the tight track cuts to the nominal track cuts are shown. Each p_T^{jet} bin is fit to a linear function to estimate the systematic uncertainty. Note the log axis in the z distribution, which makes the fit look non-linear.

6.2 Track Selection Purity

The purity of the track selection is determined by imposing a tight ghost track killing criterion and redoing the entire analysis. Nominally tracks with `ProbNNghost` < 0.5 are included in the particle flow algorithm and analyzed. To estimate the selection uncertainty, this is tightened to require that tracks pass `ProbNNghost` < 0.3 and the analysis is performed again. The difference in the tightened distributions to the nominal distributions is taken as the systematic uncertainty. The ratios of the distributions with the tight track cuts to the nominal track cuts are shown in Fig. 39. The ratios are always consistent with unity, within the large uncertainties due to the reduction in statistical precision. Since the points vary around unity within large uncertainties, a linear fit was used to estimate the systematic uncertainty at a given observable value. These fits are shown in the figure, and the resulting systematic uncertainty assigned for the track selection varies between 1-10%, depending on the bin.

6.3 Track Reconstruction Efficiency

The track reconstruction efficiency systematic uncertainty was estimated by following a procedure more closely aligned with Ref. [23]. Here, the efficiency is determined in MC and then the resulting efficiency is smoothed to give a more continuous distribution as a function of p and η. In this analysis, the track efficiency was determined bin-by-bin; thus, to estimate a systematic uncertainty on the track reconstruction efficiency, the efficiency is smoothed similarly to Ref. [23] and then the analysis is performed again with the

smoothed efficiency. The ratio of the distributions with the nominal track efficiency to the distributions determined with the smoothed efficiency is taken as a systematic uncertainty. Figure 40 shows the ratios of these distributions as a function of the hadronization variables; the figures show that in most cases the change is small. However, for some observables the difference can reach up to 10% in kinematic regions where the track efficiency would be expected to be more uncertain. For example, the differences in the distributions reach 10-15% at small z where the track efficiency is least well known due to the smaller bending radius of the tracks at small momentum. This is a conservative estimate, and regardless the measurement is statistically dominated at low z so it does not effect any conclusions to assign a conservative systematic here. The differences in the ratios are assigned as a systematic uncertainty for the track reconstruction efficiency, which is in general less than a few percent.

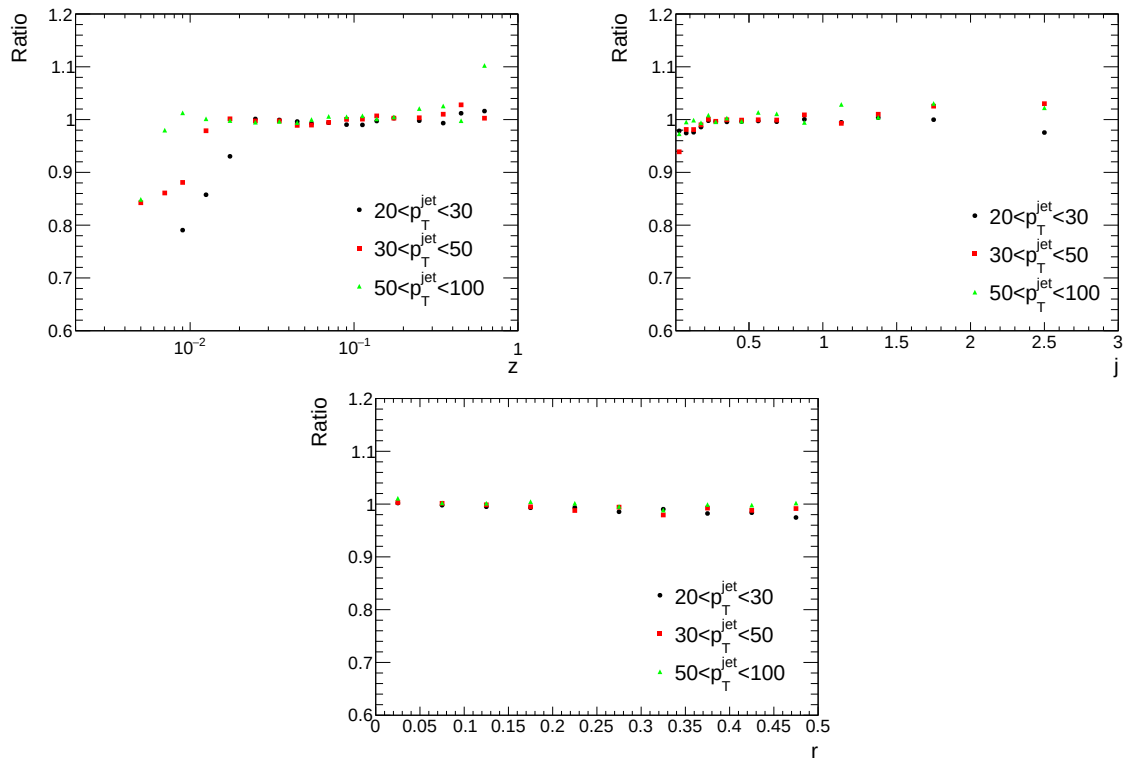

Figure 40: The ratio of the distributions determined with the nominal track efficiency to the smoothed track efficiency. The differences are taken as systematic uncertainties.

6.4 Particle Misidentification

The particle identification is largely performed by the particle flow algorithm [27]. The particle flow algorithm implements the standard LHCb cuts to select between charged hadrons and leptons. Thus, there exists the possibility that charged hadrons are mis-identified as leptons, or leptons are mis-identified as charged hadrons. To study this as a possible source of systematic uncertainty, the hadron→lepton and lepton→hadron misidentification probabilities were taken from the LHCb detector performance publication [36]. Note that since we aren't explicitly identifying the charged hadrons as pions, kaons, or protons, we are

only concerned with the probabilities of misidentifying charged hadrons as charged leptons, or vice versa.The fragmentation functions were redetermined with the mis-identification probabilities implemented, and the resulting fragmentation functions were compared with the nominal fragmentation functions. The ratios of the nominal fragmentation functions to those with the mis-identification probabilities considered are shown in Fig. 41. The difference from unity is taken as a the systematic uncertainty. Note that the expected behavior can be seen in z, since this is highly correlated to momentum. For example, the probability of mis-identification becomes larger at large z, which corresponds to large p where the $\pi \to e$ misidentification probability becomes larger.

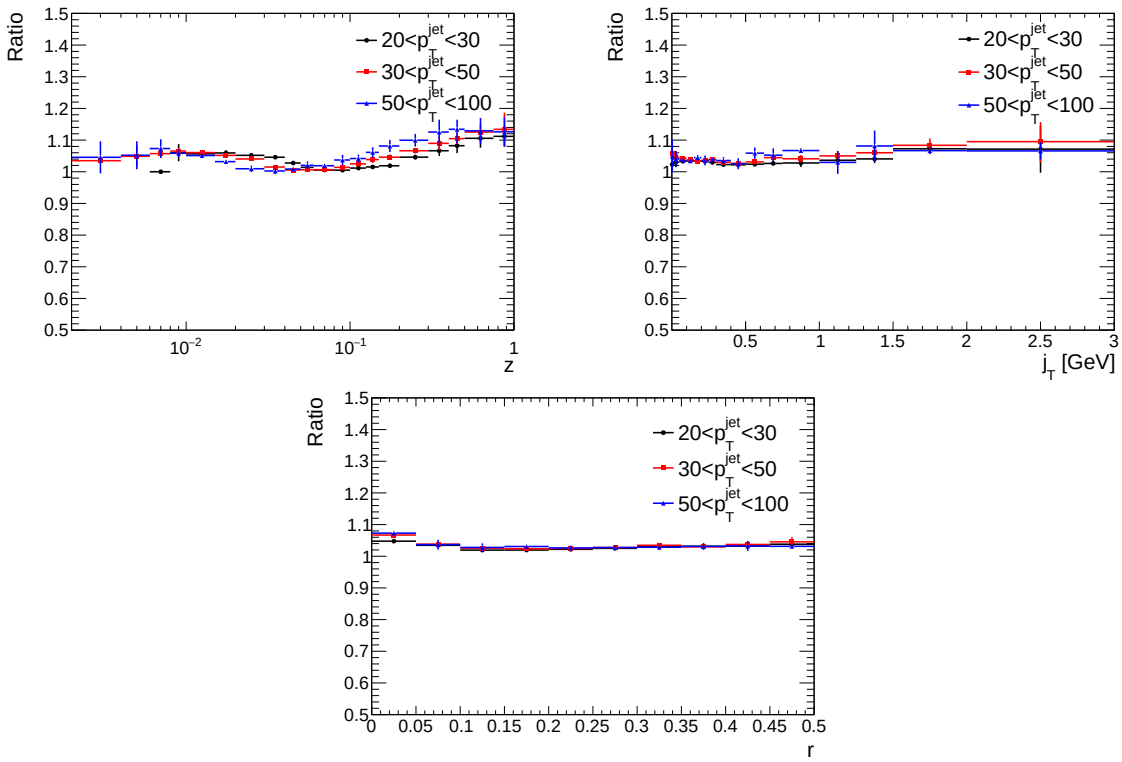

Figure 41: The ratio between the nominal fragmentation functions in this analysis and the fragmentation functions determined with a mis-ID probability included.

6.5 Jet Energy Scale

The jet energy scale (JES) was studied and validated in the Z-jet cross section analysis [21]. To determine the effect of the JES on the track-in-jet distributions, we construct new response matrices where each reconstructed jet momentum is systematically shifted by 1σ of the JES uncertainty. Similar techniques were used to evaluate this uncertainty in e.g. Refs. [18, 23]. The distributions are unfolded with the new response matrices and the difference between the nominal unfolded distributions and JES modified unfolded distributions are taken as the systematic uncertainty. This procedure is performed for each observable, and the uncertainty is assigned on a point by point basis. The level of the uncertainties are dependent on the observable; for example the JES uncertainty is larger for z than it is for r since the jet energy is directly relevant in the determination of

z, whereas for r the smearing in the JES changes the rapidity of the jet which is then propagated to r. The JES uncertainties range from 1-15% for j_T, 1-50% for z, and 1-15% for r; in general the uncertainty is less than 10% but it can rise considerably at e.g. small or large z where small changes in the jet energy can make a significant difference in the determination of z. Additionally, the uncertainty generally grows as a function of p_{T} since the JES uncertainty is larger at higher p_{T}.

6.6 Jet Energy Resolution

The effect of the jet energy resolution (JER) was evaluated by smearing the jet p_T and constructing new response matrices with which to unfold the data. The smearing was performed similarly to the Z-jet cross section analysis, where the momentum components of each jet were randomly smeared based on a Gaussian distribution of mean 1.0 and width 0.1 [21, 22]. It is important to smear each component of the jet momentum, since the various observables depend on different components of the 4 vector. The response matrices were reproduced with this random smearing, and the distributions were unfolded. Systematic uncertainties on each point are determined by the difference between the smeared unfold distributions and the nominally unfolded distributions, similarly to the JES uncertainties. The JER uncertainties vary from 1-10% in j_T, 1-5% in r, and 1-30% in z. The JER uncertainties generally decrease as a function of jet p_{T}, since the resolution improves slightly with p_{T}. Similarly to the JES uncertainties, the JER uncertainties are generally less than 10% except in extreme cases like at small and/or large z, where fluctuations in the jet momentum have a greater effect on the observable.

6.7 Unfolding

To test the response matrices, a closure test is performed by splitting up the MC data sample into two, similarly to Ref. [18]. Since the data is split into two, these are two independent data sets and the response matrices can be determined with one half and applied to the other. Recovery of the truth distributions is indicative that the response matrices are properly accounting for the bin migration effects. The response matrices were generated with the magnet down MC data and then applied to the reconstructed distributions in the magnet up MC data. Figure 42 shows the ratios of the unfolded to truth distributions when the magnet down response matrices are applied to the reconstructed magnet up distributions. The unfolding was performed with a Bayes technique iterated 7 times. The iteration choice was determined such that an additional iteration resulted in a change in the distributions of less than 0.2%. The figures show that, within the uncertainties allowed by the MC, the ratios are consistent with unity across the various kinematic variables. Each figure shows a constant fit to the data points in each jet p_{T} bin, and conservatively the largest deviation from unity is taken as a systematic uncertainty to account for the unfolding procedure. For each observable the largest deviation is 2%, so this is taken as a systematic uncertainty to account for the unfolding procedure. Adding or subtracting an additional iteration in the Bayesian unfolding resulted in a difference in the distributions of $\mathcal{O}(0.1\%)$, so any uncertainty in the choice of iteration number is neglected.

An additional closure test was performed which used different starting truth distributions to test the assumed prior. The MC data was split by Z boson p_{T} instead of by

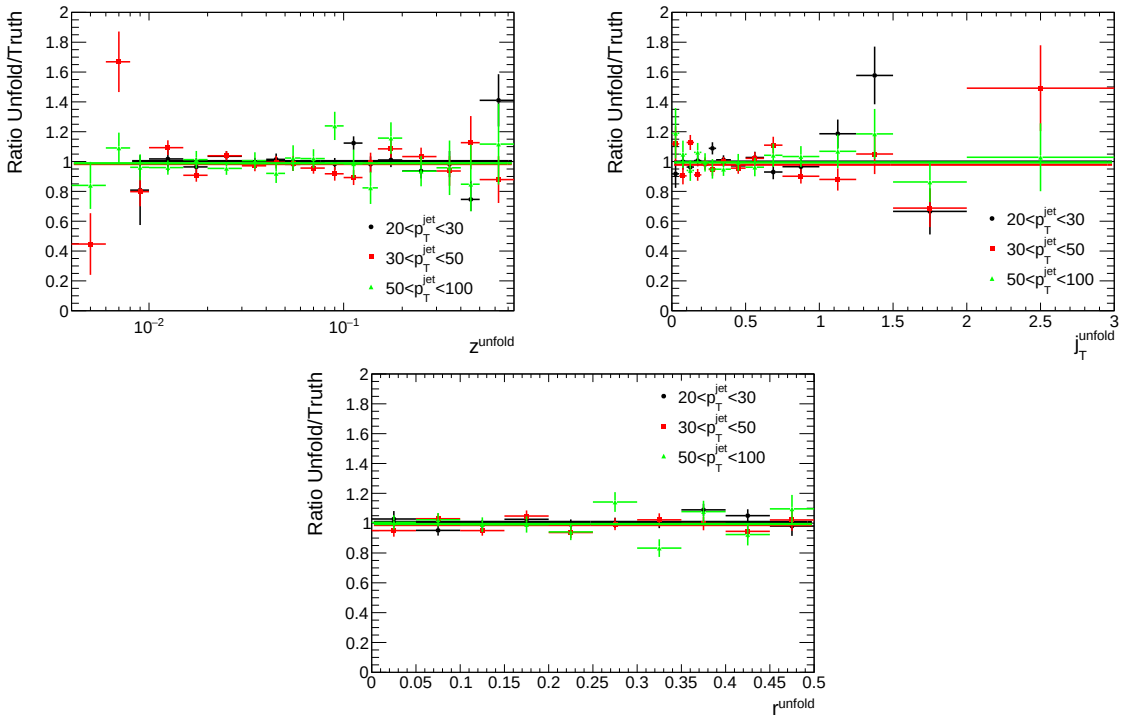

Figure 42: Closure tests performed as described in the text. The figures show the ratio between the unfolded and truth distributions and are consistent with unity within the uncertainties.

magnet down or up configurations and used to generate response matrices. Since there is a strong correlation between the Z boson p_{T} and the jet p_{T}, the MC data was split in a way such that the method could be tested in as independent a way as possible. The response matrices were generated with only the events where the Z p_{T} was greater than 40 GeV. These response matrices were applied to the MC with Z p_{T} less than 40 GeV. The highest jet p_{T} bin is unable to be tested with this method due to the statistical precision of the sample; there are very few jets with $p_{\mathrm{T}} > 50$ GeV for Z bosons with $p_{\mathrm{T}} < 40$ GeV. There is also an auto correlation with the $30 < p_T^{jet} < 50$ GeV bin in this test, so we perform the closure test on the $20 < p_T^{jet} < 30$ GeV bin so that it is completely independent. The results of the closure test are shown in Fig. 43, and show that the results are consistent with unity within the 2% systematic uncertainty assigned from the previous closure test.

6.8 Summary

The following tables show a summary of the various systematic uncertainties. Each uncertainty also has a comment to denote how it is applied, e.g. whether or not it is an overall normalization uncertainty or is point-to-point. Note that in general the quadrature sum of the systematic uncertainties bin-by-bin is less than 10%, and only in extreme cases (e.g. at high z) do the systematic uncertainties become as large as what is indicated in the summary table. The figures following the tables also show visual representations of the point by point uncertainties and their relative amounts.

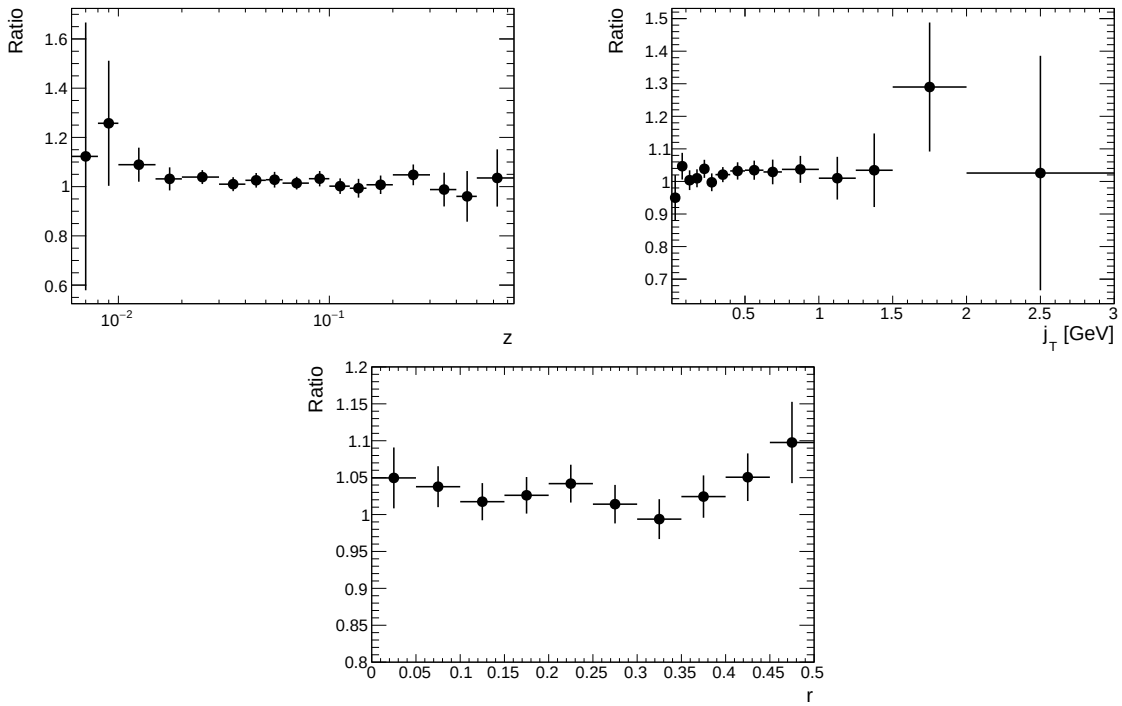

Figure 43: A second closure test, described in the text, performed by splitting the data in half by Z boson p_{T}.

Table 2: A summary of the normalization systematic uncertainties assigned, shown as percentages.

Systematic Source	Relative Uncertainty
Muon reconstruction efficiencies	0.7%
Jet reconstruction efficiencies	1.9%
Z-jet Selection purity	1.7%
Quadrature Sum	2.7%

Table 3: A summary of the point-to-point systematic uncertainties assigned, shown as percentages. Note that the overall systematic assigned is highly dependent on the observable and the bin, as can be seen in the distributions in Section 7.

Systematic Source	Relative Uncertainty
Track Selection	1-10%
Track Efficiencies	1-20%
JES	1-50%
JER	1-30%
PID mis-identification	1-10%
Unfolding	2%
Quadrature Sum	4-70%

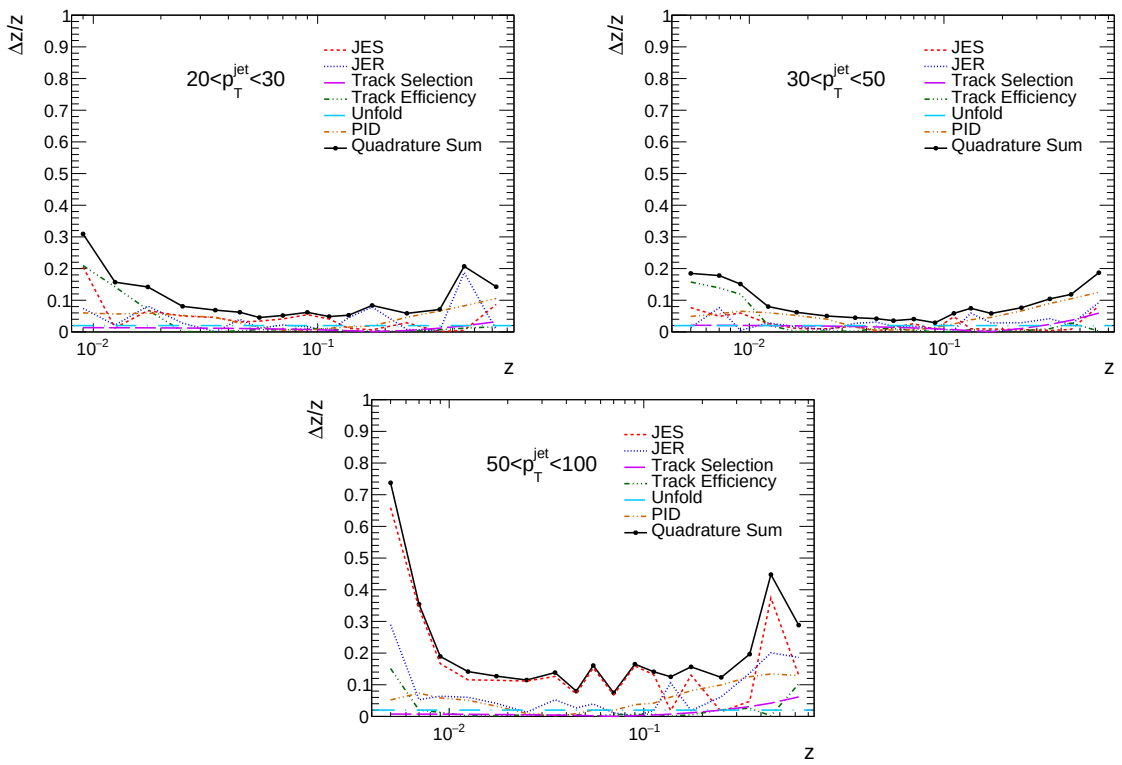

Figure 44: A summary of the point-to-point systematic uncertainties on z, in each of the jet p_{T} bins. The individual contributions are shown as various lines, and the quadrature sum is shown as the black solid line and points.

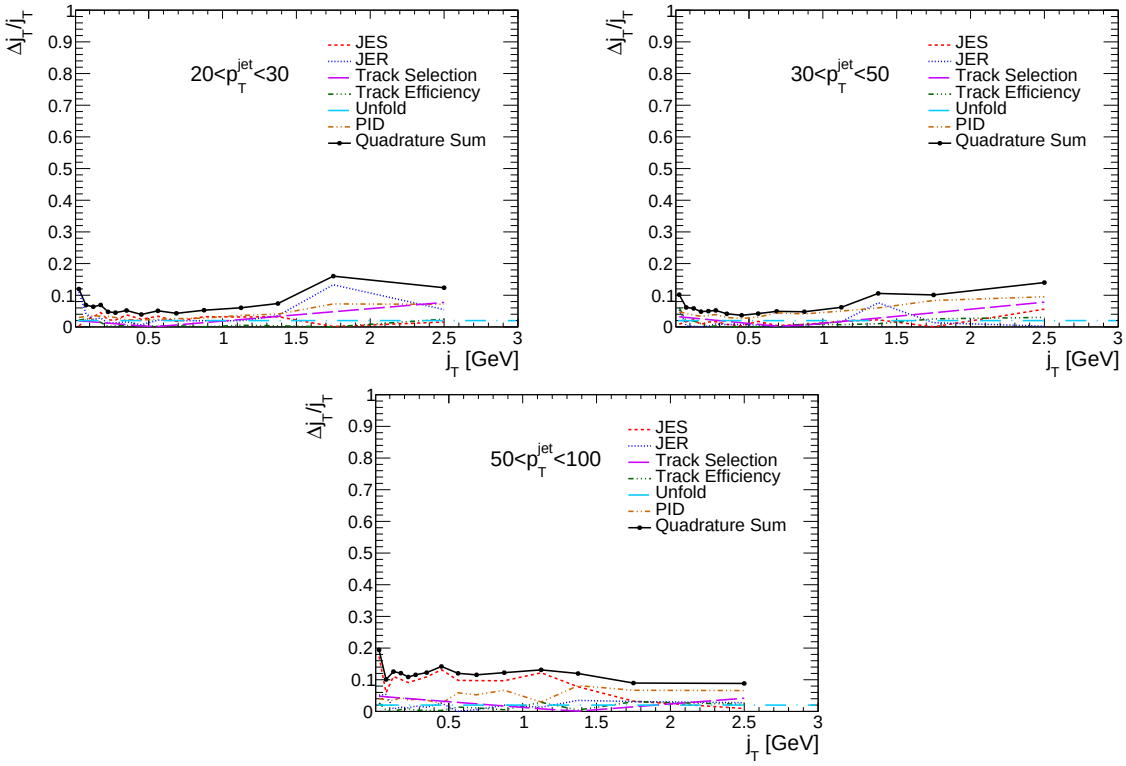

Figure 45: A summary of the point-to-point systematic uncertainties on j_T, in each of the jet p_{T} bins. The individual contributions are shown as various lines, and the quadrature sum is shown as the black solid line and points.

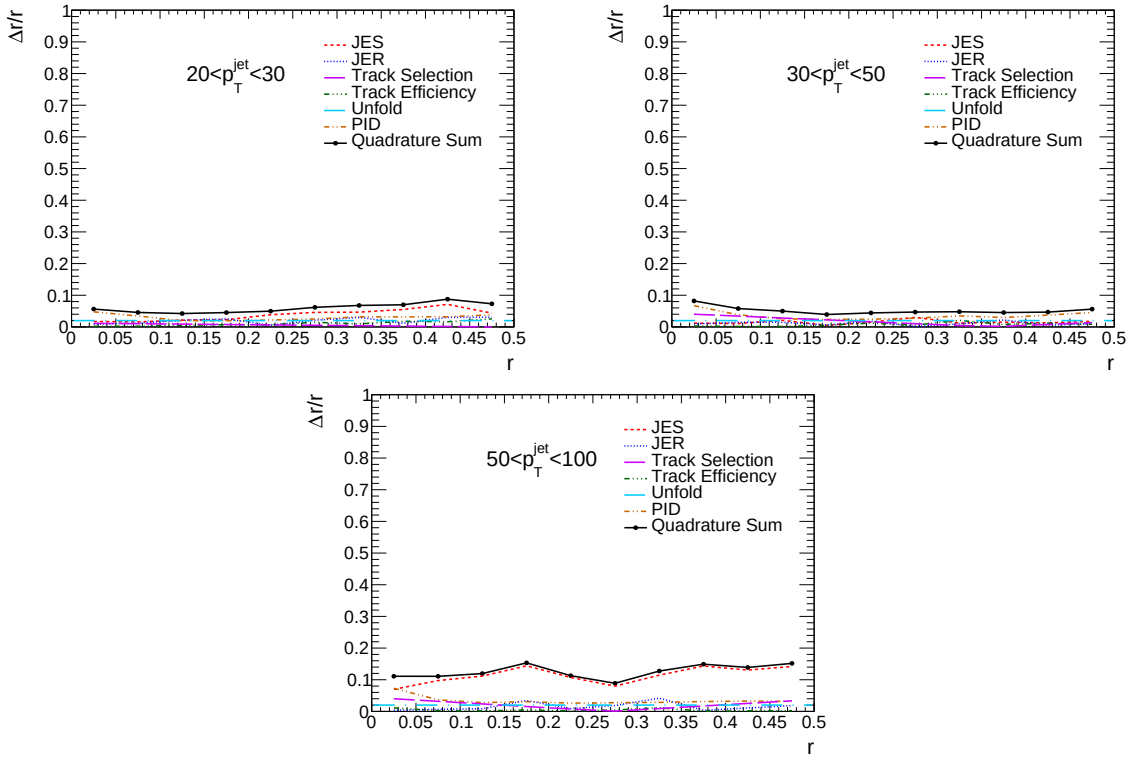

Figure 46: A summary of the point-to-point systematic uncertainties on r, in each of the jet p_{T} bins. The individual contributions are shown as various lines, and the quadrature sum is shown as the black solid line and points.

7 Results

The final results are the unfolded distributions shown with all of the systematic uncertainties from Section 6 added in quadrature. These are shown in Fig. 47 as a function of p_T^{jet} and the various hadronization variables. The distributions are, in almost all cases, statistically limited rather than systematics limited. They also show general features expected from other measurements; for example, the z distributions are independent of p_T^{jet} at high z which has been seen in central rapidity inclusive jet measurements at ATLAS [14] and ALICE [15, 16]. As another example, the j_T spectra at large j_T become harder, or more power law like, which is generally expected from TMD evolution expectations [1].

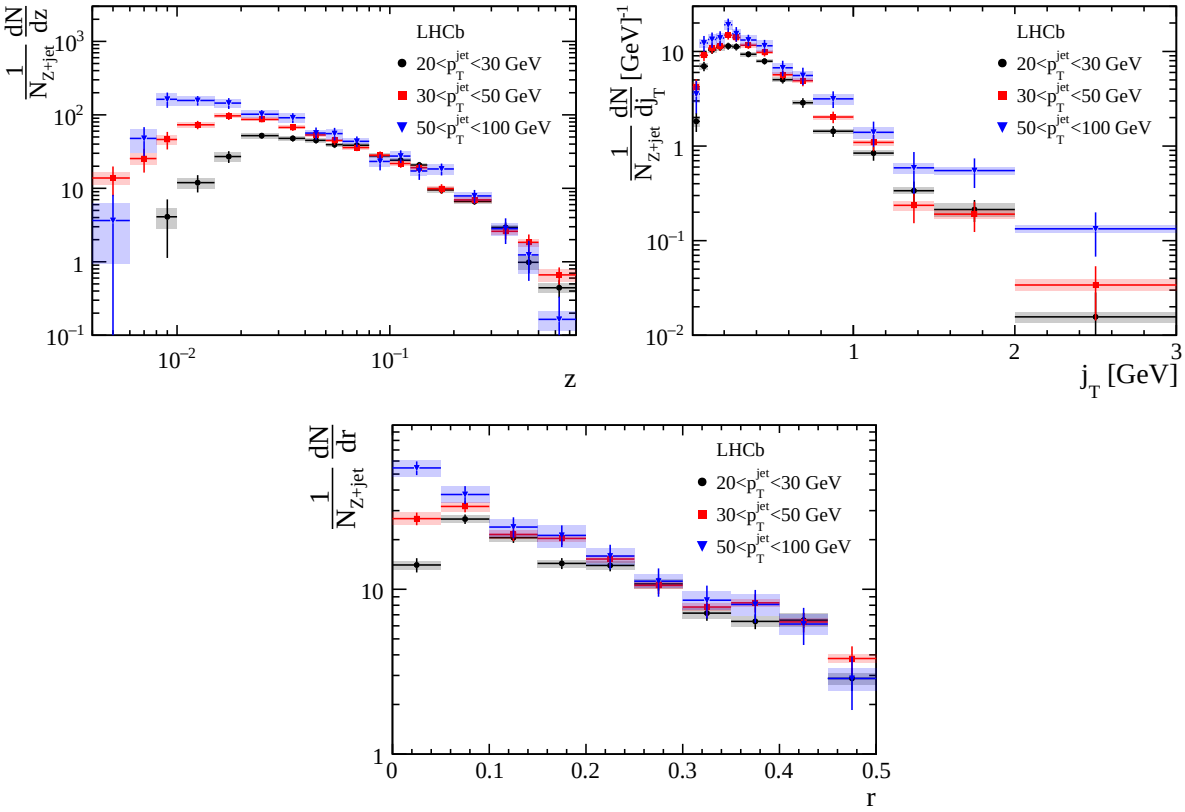

Figure 47: The final hadronization distributions are shown as a function of p_T^{jet} and the various observables, where all systematic uncertainties are combined in quadrature.

To compare the results to a MC generator, the measured distributions are compared to the Pythia 8 MC distributions. The MC distributions were constructed with the same truth level cuts as the data are, shown in Table 1. The data and MC distributions are shown in Fig. 48, where the distributions are scaled by an arbitrary factor so that they can all be viewed on the same plot. To facilitate the comparison, the ratios of the MC to data are taken and the results are shown in Figs. 49, 50, and 51, where the red points are the MC-to-data ratios and the black points and black shaded rectangles show the statistical and systematic uncertainties, respectively.

In general the comparisons to the MC show two features. The first feature is that the MC systematically under-predicts high z hadrons within Z tagged jets at forward rapidities. This is reflected in the MC-to-data ratio less than unity at high z, and is also

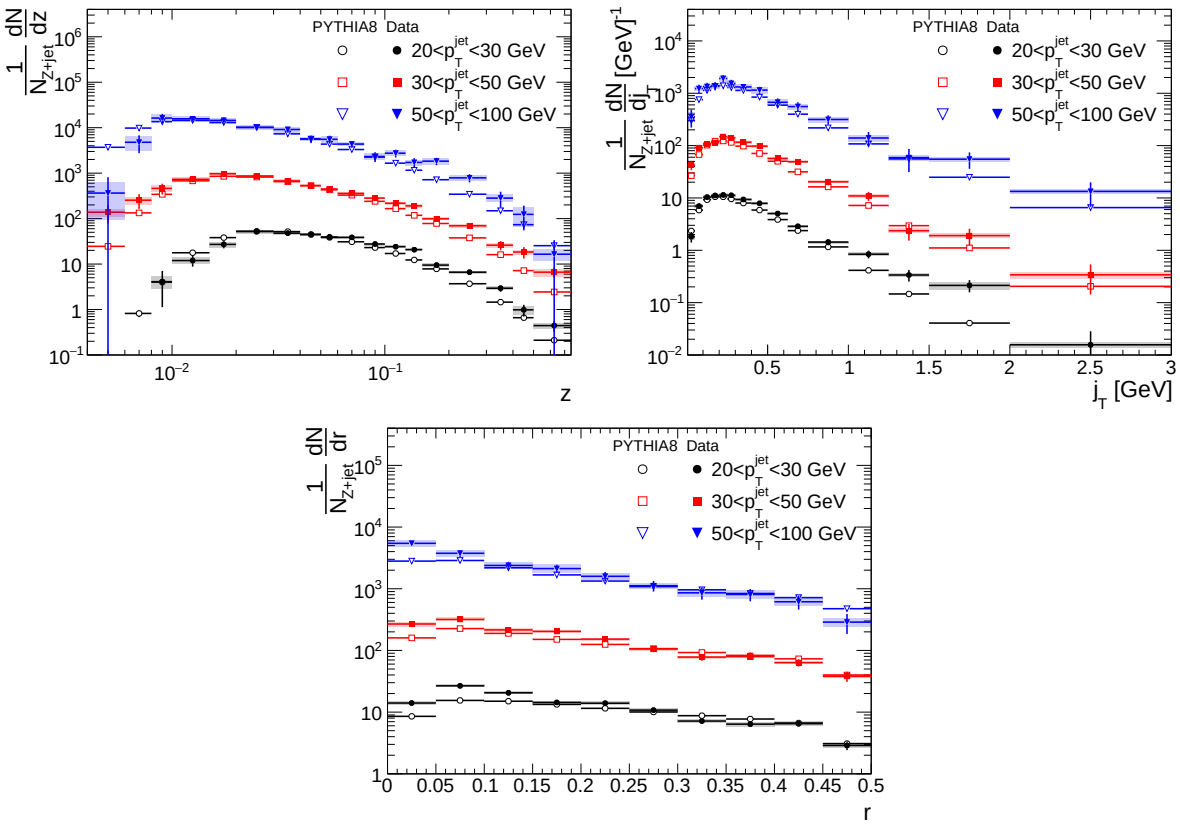

Figure 48: Comparisons between the distributions in data and PYTHIA 8 MC are shown. The distributions for $p_T^{jet} > 30$ GeV are scaled by an arbitrary factor so that they can be viewed on the same plot.

reflected in r where the MC distributions under-predict the data at very small r. This indicates that the fragmentation mechanism in the MC is in general under producing high z hadrons which form the core of the anti-k$_T$ jet. The second feature is that the MC almost always under-predicts the j_T distribution across all j_T, and especially at large j_T. This under-prediction, and the consistency between the data and MC, generally becomes better at high p_T^{jet}, although the uncertainties also become larger here. It is worth reiterating that this is the first measurement of hadronization within jets at the LHC at very forward rapidities, in addition to being the first measurement of hadronization within Z tagged jets at the LHC; therefore, excellent agreement between the data and MC should not necessarily be expected. It is also worth mentioning that this particular PYTHIA 8 MC was generated in 2013, and since then much more data has become available for tuning MC generators.

We also plan on comparing the results to NLO calculations within the SCET framework from the QCD group at University of California Los Angeles. We have requested these calculations and they are currently being worked on; details of the theoretical background for the calculations can be found in Refs. [6, 7].

Part of the motivation of this study is to also provide more data for TMD FF fits, which are used to constrain the three dimensional nonperturbative fragmentation functions. The j_T distributions themselves can do this; however, it is also useful to extract momentum widths which are sensitive to nonperturbative transverse momentum from fits to the

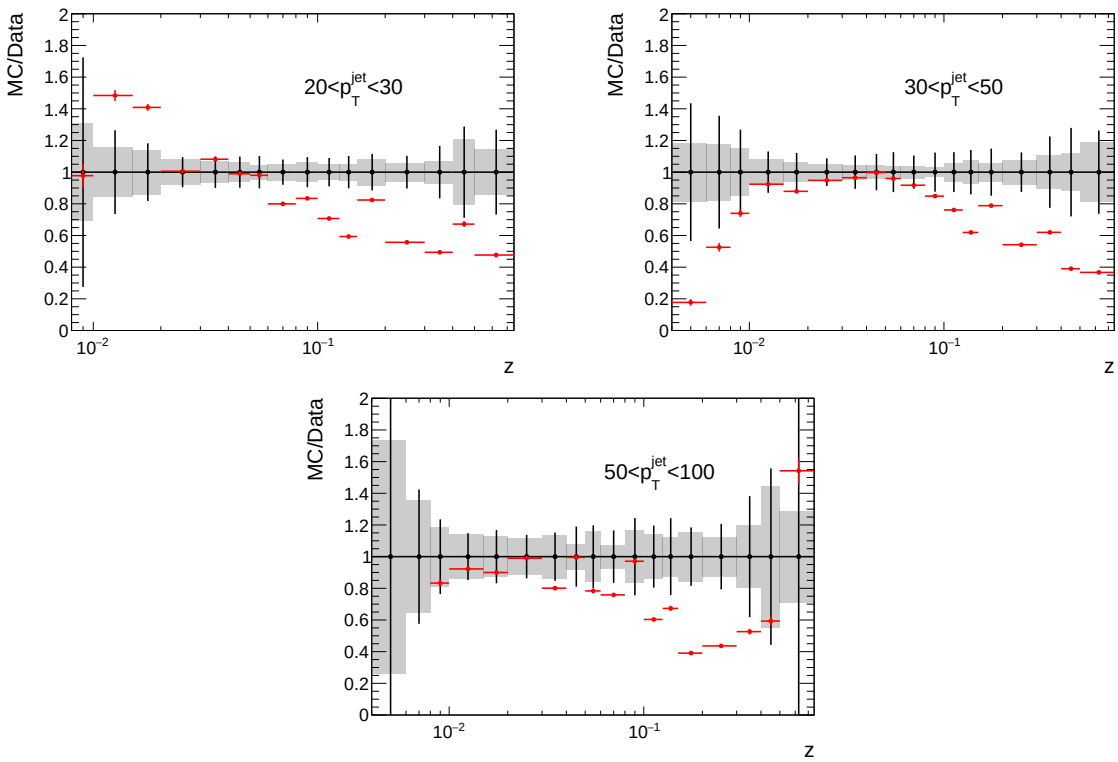

Figure 49: Ratios of the distributions of z in MC to data are shown as a function of p_T^{jet}. The statistical uncertainties of the data are shown as bars and the systematic uncertainties are shown as shaded boxes in each bin.

distributions. Future phenomenological analyses can then use these data to, for example, constrain TMD evolution calculations, which currently vary drastically depending on assumptions and inputs for the nonperturbative contributions [37]. These nonperturbative contributions are understood to be universal, in that they are the same for PDFs, FFs, and unpolarized or polarized process, with the only difference coming between quark and gluon contributions (see e.g. Ref. [38] and Refs. within).

To measure nonperturbative TMD widths, the j_T distributions were fit to a Gaussian times power law function to model the nonperturbative and perturbative behavior of the distributions. The fits to the data are shown in Fig. 52, where the fit is performed in the range of $[0, 1.5]$ GeV which was chosen as it gives the best χ^2/NDF for the three fits. At larger j_T, more detailed NLO effects become relevant and these are not well described by a basic power law function. The width of the Gaussian part of the function is extracted and plotted as a function of p_T^{jet} in Fig. 53. Systematic uncertainties on the fit function and determination of $\langle j_T \rangle$ are estimated by adjusting the fit region by one bin and taking the absolute value of the difference of the resulting fit parameters. The $\langle j_T \rangle$ values are sensitive to nonperturbative contributions only, and display a rising trend with p_T^{jet}, within somewhat large uncertainties, which is expected by TMD evolution and has been measured at, for example, CDF [39].

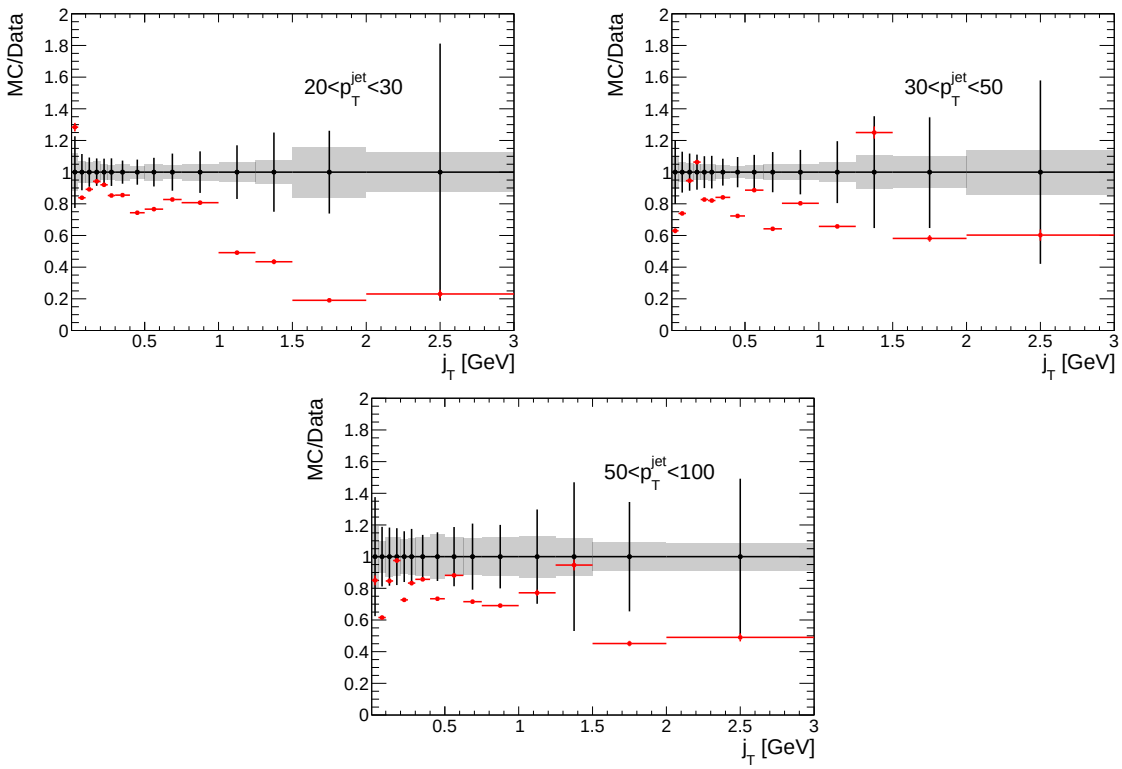

Figure 50: Ratios of the distributions of j_T in MC to data are shown as a function of p_T^{jet}. The statistical uncertainties of the data are shown as bars and the systematic uncertainties are shown as shaded boxes in each bin.

8 Summary

In summary, the fragmentation distributions of charged hadrons within jets in conjunction with a Z boson have been measured. The measurement is made with the 2012 data collected by LHCb at $\sqrt{s} = 8$ TeV, corresponding to an integrated luminosity of approximately 2 fb $^{-1}$. These measurements constitute the first measurement of charged hadron jet substructure at LHCb, and also are the first measurement of charged hadrons in jets opposite a Z boson at the LHC. Similar measurements have been made at ATLAS [14, 18, 19, 40] and CMS [17, 41] with a stronger focus on understanding cold and/or hot nuclear matter effects in the presence of a heavy nucleus.

The longitudinal, transverse, and radial distributions are measured for $R = 0.5$ anti-k$_T$ jets with $p_{\mathrm{T}} > 20$ GeV and $2.5 < \eta < 4$, in several p_{T} bins. The Z bosons are measured via their dimuon decay channel, where both muons are required to satisfy $2 < \eta < 4.5$ and the invariant mass of the pair satisfies $60 < M_{\mu\mu} < 120$ GeV. No constraint is placed on the Z p_{T} so that the statistical precision can be used for making the measurement in bins of p_T^{jet}; however, the various cuts at analysis level result in the Z boson having $0 < p_{\mathrm{T}} < 100$ GeV. The Z-jet pair are required to be nearly back-to-back in azimuth such that $\Delta\phi_{Z-jet} > 7\pi/8$, which tags events that have a more $2 \to 2$ hard scattering topology. Charged hadrons are required to have $4 < p < 1000$ GeV and $p_{\mathrm{T}} > 250$ MeV, and are additionally required to have $\Delta R < 0.5$.

The unfolded distributions are compared to Z-jet Pythia 8 events that satisfy the same truth criteria. The comparison to Pythia shows that the generator in general under

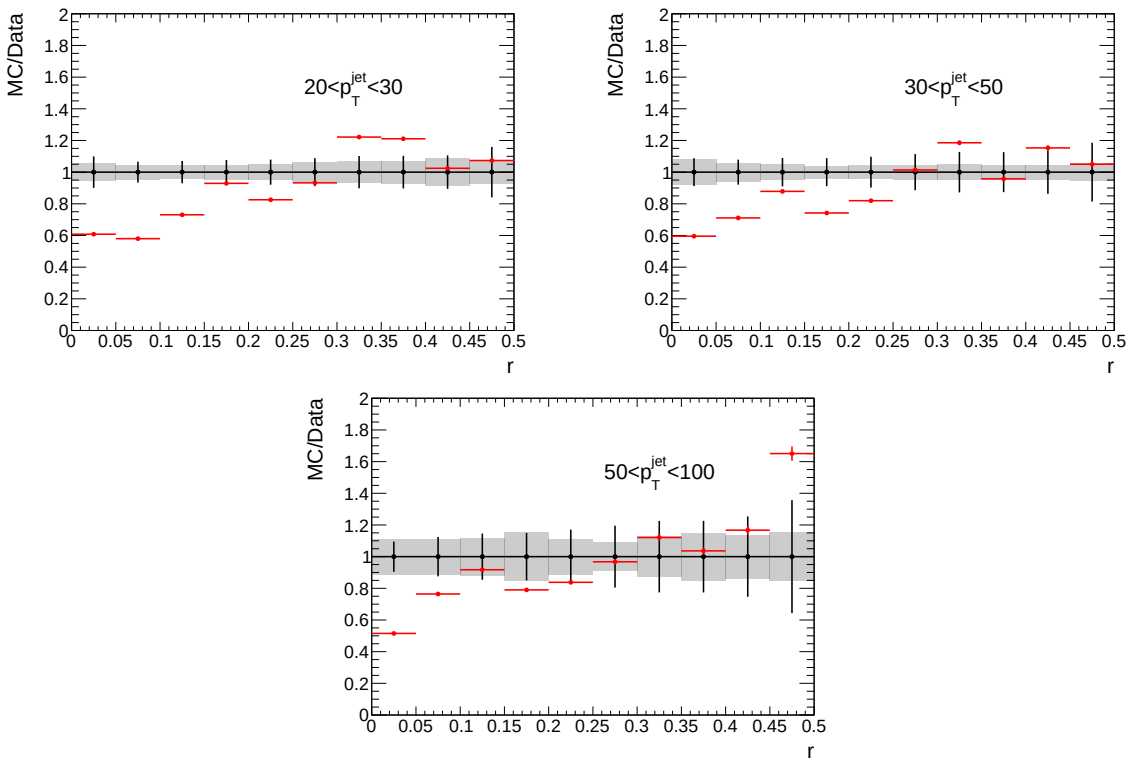

Figure 51: Ratios of the distributions of r in MC to data are shown as a function of p_T^{jet}. The statistical uncertainties of the data are shown as bars and the systematic uncertainties are shown as shaded boxes in each bin.

predicts high momentum hadrons within jets, which manifests itself as a lack of hadrons at high z and at small r. Given that these are the first jet substructure results at very forward rapidities from the LHC, these results will contribute to MC event generator nonperturbative fragmentation tuning in the future. The results are also used to extract quantities sensitive to TMD fragmentation and TMD evolution effects, which require data to constrain certain nonperturbative inputs. The nonperturbative transverse momentum widths increase with p_T^{jet}, which is qualitatively expected from TMD evolution.

We note also that this analysis is intended to lay a foundation for a broader research program investigating jet substructure at LHCb. For example, future studies can include utilizing the excellent particle identification detectors at LHCb, or the already demonstrated capabilities of heavy quark tagging through secondary vertex identification. Ideally, one can use the LHCb apparatus to carefully identify the final-state fragmenting quark and the corresponding hadrons within the resulting jet, which will allow for robust correlations to be drawn between the fragmenting quark and the final-state (particle identified) hadrons. This opens an entirely new research path into studying hadronization, fragmentation, and generically jet substructure, where a particular jet can have many of its constituents identified in addition to more careful identification of the flavor (light or heavy) of the parton which produced the jet.

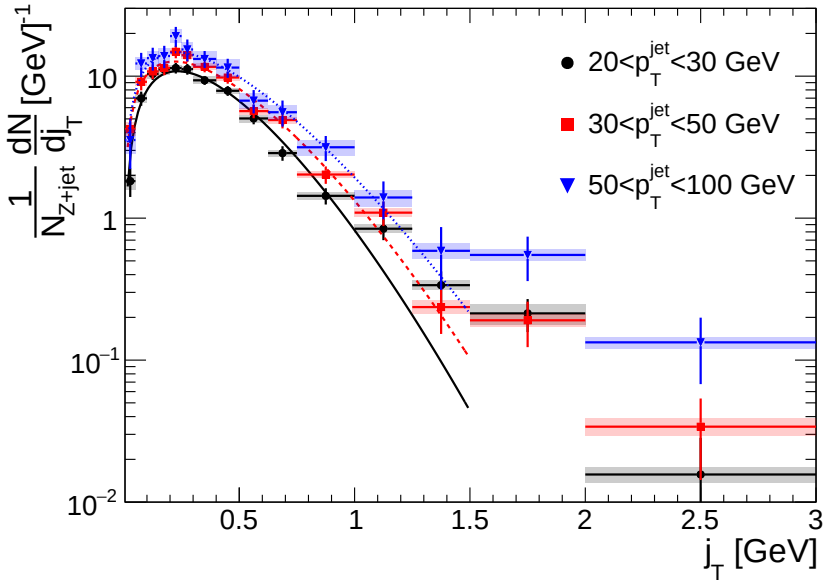

Figure 52: The j_T distributions are shown with fits as described in the text. The fits are used to extract the nonperturbative momentum widths of the distribution, which are sensitive to nonperturbative TMD evolution contributions.

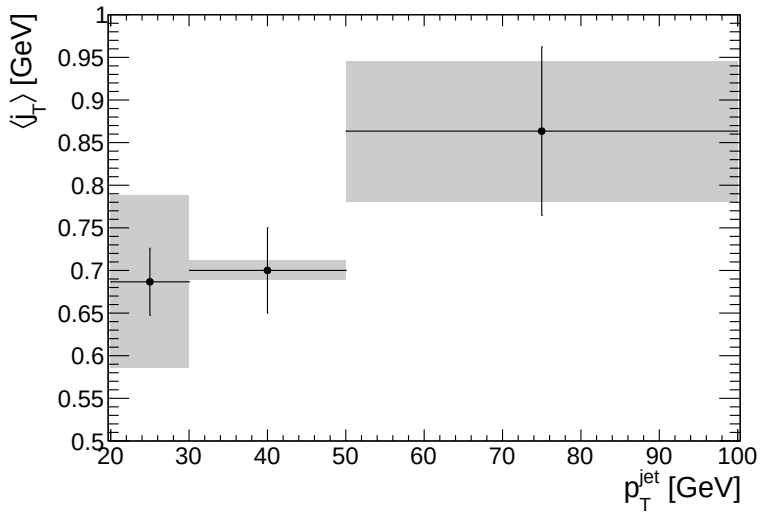

Figure 53: The nonperturbative $\langle j_T \rangle$ is shown as extracted from the fit function.

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A Data Samples and Paths

The MC path used for this analysis is with code 42112022 at the path `/MC/2012/42112022/Beam4000GeV-2012-MagDown-Nu2.5-Pythia8/Sim08b/Digi13/Trig0x409f0045/Reco14a/Stripping20NoPrescalingFlagged/ALLSTREAMS.DST`, where the magnet polarity can be either 'Up' or 'Down.'

The corresponding data path used for this analysis is the stripping 20 data at the path `/LHCb/Collision12/90000000/Beam4000GeV-VeloClosed-MagDown/Real Data/Reco14/Stripping20/WG-EW-Selection/Merging/ZMUMULINE.DST`, where again the magnet polarity can be either 'Up' or 'Down.'

B Data Tables

The following tables give the final unfolded distributions, where all systematic uncertainties are added in quadrature.

Table 4: The total efficiency corrected N^{true}_{Z+jet} counts, given in three jet p_{T} bins. These values account for the normalization of the distributions.

Jet p_{T}	Counts	(stat)	(syst)
20-30	758	31	20
30-50	515	25	14
50-100	194	15	5

Table 5: The unfolded charged hadron-in-jet distributions for jet p_{T} $20 < p_{\mathrm{T}} < 30$ GeV.

z	$1/N_{jet}dN/dz$	(stat)	(syst)
0.008-0.01	4.10	2.97	1.27
0.01-0.015	11.9	3.2	1.9
0.015-0.02	27.1	4.9	3.9
0.02-0.03	52.1	5.0	4.4
0.03-0.04	47.9	4.8	3.5
0.04-0.05	45.0	4.4	3.0
0.05-0.06	39.3	4.0	2.1
0.06-0.08	38.6	3.1	2.3
0.08-0.1	27.6	2.6	1.9
0.1-0.125	24.1	2.2	1.3
0.125-0.15	20.7	2.1	1.2
0.15-0.2	9.50	1.10	0.83
0.2-0.3	6.63	0.68	0.43
0.3-0.4	2.94	0.49	0.22
0.4-0.5	0.983	0.283	0.205
0.5-0.75	0.443	0.119	0.065

Table 6: The unfolded charged hadron-in-jet distributions for jet p_{T} $30 < p_{\mathrm{T}} < 50$ GeV.

z	$1/N_{jet}dN/dz$	(stat)	(syst)
0.004-0.006	13.8	6.01	2.58
0.006-0.008	25.4	9.1	4.6
0.008-0.01	46.2	12.4	7.1
0.01-0.015	73.2	9.6	6.2
0.015-0.02	96.9	11.7	6.6
0.02-0.03	86.7	7.6	4.9
0.03-0.04	67.4	7.1	3.5
0.04-0.05	53.3	6.1	2.7
0.05-0.06	44.8	5.7	2.0
0.06-0.08	35.8	3.7	1.7
0.08-0.1	28.3	3.5	1.1
0.1-0.125	21.8	2.7	1.4
0.125-0.15	19.1	2.7	1.5
0.15-0.2	9.90	1.47	0.64
0.2-0.3	6.96	0.87	0.56
0.3-0.4	2.60	0.59	0.28
0.4-0.5	1.84	0.52	0.22
0.5-0.75	0.663	0.175	0.125

Table 7: The unfolded charged hadron-in-jet distributions for jet p_{T} $50 < p_{\mathrm{T}} < 100$ GeV.

z	$1/N_{jet}dN/dz$	(stat)	(syst)
0.004-0.006	3.64	4.51	2.69
0.006-0.008	47.8	20.3	17.0
0.008-0.01	163	38	31
0.01-0.015	157	23	22
0.015-0.02	145	24	19
0.02-0.03	102	14	12
0.03-0.04	91.3	13.9	12.9
0.04-0.05	56.5	10.7	4.8
0.05-0.06	55.6	11.0	9.1
0.06-0.08	43.6	7.2	3.5
0.08-0.1	23.2	5.7	3.9
0.1-0.125	27.5	5.4	3.9
0.125-0.15	17.2	4.2	2.2
0.15-0.2	18.3	3.4	2.9
0.2-0.3	7.88	1.63	0.99
0.3-0.4	2.83	1.08	0.56
0.4-0.5	1.24	0.69	0.56
0.5-0.75	0.165	0.170	0.048

Table 8: The unfolded charged hadron-in-jet distributions for jet p_{T} $20 < p_{\mathrm{T}} < 30$ GeV.

j_T	$1/N_{jet}dN/dj_T$	(stat)	(syst)
0.0-0.05	1.82	0.41	0.22
0.05-0.1	6.98	0.80	0.52
0.1-0.15	10.4	1.0	0.7
0.15-0.2	11.2	1.0	0.8
0.2-0.25	11.4	1.0	0.6
0.25-0.3	11.2	1.0	0.6
0.3-0.4	9.35	0.68	0.55
0.4-0.5	7.89	0.63	0.38
0.5-0.625	5.04	0.46	0.29
0.625-0.75	2.87	0.34	0.15
0.75-1	1.44	0.19	0.09
1-1.25	0.842	0.140	0.056
1.25-1.5	0.337	0.084	0.027
1.5-2	0.213	0.056	0.035
2-3	0.015	0.013	0.002

Table 9: The unfolded charged hadron-in-jet distributions for jet p_{T} $30 < p_{\mathrm{T}} < 50$ GeV.

j_T	$1/N_{jet}dN/dj_T$	(stat)	(syst)
0.0-0.05	4.24	0.85	0.45
0.05-0.1	9.12	1.20	0.61
0.1-0.15	10.8	1.3	0.7
0.15-0.2	11.4	1.3	0.6
0.2-0.25	14.9	1.5	0.8
0.25-0.3	14.1	1.4	0.8
0.3-0.4	11.6	1.0	0.6
0.4-0.5	9.79	0.94	0.45
0.5-0.625	5.67	0.62	0.28
0.625-0.75	4.91	0.62	0.28
0.75-1	2.03	0.28	0.11
1-1.25	1.09	0.21	0.07
1.25-1.5	0.236	0.083	0.026
1.5-2	0.191	0.067	0.020
2-3	0.034	0.020	0.005

Table 10: The unfolded charged hadron-in-jet distributions for jet p_{T} $50 < p_{\mathrm{T}} < 100$ GeV.

j_T	$1/N_{jet}dN/dj_T$	(stat)	(syst)
0.0-0.05	3.55	1.30	0.70
0.05-0.1	12.3	2.3	1.3
0.1-0.15	13.4	2.5	1.7
0.15-0.2	13.9	2.5	1.7
0.2-0.25	19.2	3.1	2.1
0.25-0.3	15.4	2.7	1.8
0.3-0.4	13.2	1.8	1.7
0.4-0.5	11.5	1.8	1.7
0.5-0.625	6.72	1.30	0.83
0.625-0.75	5.57	1.20	0.66
0.75-1	3.16	0.63	0.40
1-1.25	1.40	0.42	0.19
1.25-1.5	0.589	0.280	0.072
1.5-2	0.550	0.190	0.051
2-3	0.133	0.066	0.012

Table 11: The unfolded charged hadron-in-jet distributions for jet p_{T} $20 < p_{\mathrm{T}} < 30$ GeV.

r	$1/N_{jet}dN/dr$	(stat)	(syst)
0-0.05	14.0	1.4	0.9
0.05-0.1	26.7	1.8	1.4
0.1-0.15	20.6	1.4	1.0
0.15-0.2	14.4	1.1	0.7
0.2-0.25	13.9	1.1	0.8
0.25-0.3	10.8	0.9	0.7
0.3-0.35	7.17	0.73	0.52
0.35-0.4	6.38	0.66	0.48
0.4-0.45	6.50	0.69	0.60
0.45-0.5	2.88	0.46	0.22

Table 12: The unfolded charged hadron-in-jet distributions for jet p_{T} $30 < p_{\mathrm{T}} < 50$ GeV.

r	$1/N_{jet}dN/dr$	(stat)	(syst)
0-0.05	26.8	2.4	2.3
0.05-0.1	31.8	2.5	2.0
0.1-0.15	21.5	1.9	1.2
0.15-0.2	20.3	1.8	1.0
0.2-0.25	15.3	1.5	0.8
0.25-0.3	10.6	1.2	0.6
0.3-0.35	7.80	1.00	0.43
0.35-0.4	8.27	1.04	0.44
0.4-0.45	6.34	0.87	0.34
0.45-0.5	3.80	0.70	0.24

Table 13: The unfolded charged hadron-in-jet distributions for jet p_{T} $50 < p_{\mathrm{T}} < 100$ GeV.

r	$1/N_{jet}dN/dr$	(stat)	(syst)
0-0.05	54.5	5.2	6.2
0.05-0.1	37.6	4.7	4.3
0.1-0.15	23.9	3.5	2.9
0.15-0.2	21.2	3.2	3.3
0.2-0.25	15.9	2.7	1.8
0.25-0.3	11.2	2.2	1.0
0.3-0.35	8.58	1.94	1.12
0.35-0.4	8.08	1.83	1.23
0.4-0.45	6.15	1.56	0.87
0.45-0.5	2.88	1.03	0.44

Because unfolding techniques may/can change and improve with time, the efficiency corrected charged hadron distributions are included here in addition to the distributions after unfolding (shown above). The hope is that, by including these and making them available, future studies may revisit these data points when and/or if unfolding procedures improve in the future. The systematic uncertainties are not assigned to these distributions, since many of the systematic uncertainties involve the unfolding procedure.

Table 14: The efficiency corrected charged hadron-in-jet distributions for jet p_{T} $20 < p_{\mathrm{T}} < 30$ GeV. These values are before the unfolding procedure is performed.

z	dN/dz	(stat)
0.008-0.01	3.3e+03	1.48e+03
0.01-0.015	1.14e+04	1.74e+03
0.015-0.02	2.52e+04	2.65e+03
0.02-0.03	4.27e+04	2.47e+03
0.03-0.04	4.06e+04	2.43e+03
0.04-0.05	3.66e+04	2.34e+03
0.05-0.06	3.04e+04	2.15e+03
0.06-0.08	2.9e+04	1.52e+03
0.08-0.1	2.2e+04	1.34e+03
0.1-0.125	1.78e+04	1.1e+03
0.125-0.15	1.55e+04	1.07e+03
0.15-0.2	8.19e+03	574
0.2-0.3	5.03e+03	329
0.3-0.4	2.2e+03	232
0.4-0.5	843	148
0.5-0.75	379	65.6

Table 15: The efficiency corrected charged hadron-in-jet distributions for jet p_{T} $20 < p_{\mathrm{T}} < 30$ GeV. These values are before the unfolding procedure is performed.

j_T [GeV]	dN/dj_T [GeV]$^{-1}$	(stat)
0-0.05	1.44e+03	210
0.05-0.1	5.38e+03	409
0.1-0.15	8.07e+03	503
0.15-0.2	9.12e+03	533
0.2-0.25	9.49e+03	555
0.25-0.3	9.02e+03	558
0.3-0.4	7.56e+03	359
0.4-0.5	6.2e+03	333
0.5-0.625	3.97e+03	245
0.625-0.75	2.45e+03	194
0.75-1	1.22e+03	101
1-1.25	564	69.6
1.25-1.5	218	42.6
1.5-2	94.2	21.1
2-3	3.83	3.83

Table 16: The efficiency corrected charged hadron-in-jet distributions for jet p_{T} $20 < p_{\mathrm{T}} < 30$ GeV. These values are before the unfolding procedure is performed.

r	dN/dr	(stat)
0-0.05	1.01e+04	651
0.05-0.1	1.8e+04	850
0.1-0.15	1.58e+04	758
0.15-0.2	1.19e+04	632
0.2-0.25	1.13e+04	605
0.25-0.3	9.08e+03	538
0.3-0.35	6.87e+03	460
0.35-0.4	6.02e+03	428
0.4-0.45	5.74e+03	411
0.45-0.5	2.31e+03	260

Table 17: The efficiency corrected charged hadron-in-jet distributions for jet p_{T} $30 < p_{\mathrm{T}} < 50$ GeV. These values are before the unfolding procedure is performed.

z	dN/dz	(stat)
0.004-0.006	4.55e+03	1.72e+03
0.006-0.008	1.23e+04	2.95e+03
0.008-0.01	2.03e+04	3.72e+03
0.01-0.015	3.66e+04	3.2e+03
0.015-0.02	4.8e+04	3.69e+03
0.02-0.03	4.35e+04	2.52e+03
0.03-0.04	3.39e+04	2.29e+03
0.04-0.05	2.86e+04	2.11e+03
0.05-0.06	2.35e+04	1.95e+03
0.06-0.08	1.9e+04	1.28e+03
0.08-0.1	1.51e+04	1.19e+03
0.1-0.125	1.12e+04	950
0.125-0.15	9.49e+03	893
0.15-0.2	5.55e+03	507
0.2-0.3	3.44e+03	294
0.3-0.4	1.44e+03	198
0.4-0.5	905	161
0.5-0.75	321	61.9

Table 18: The efficiency corrected charged hadron-in-jet distributions for jet p_{T} $30 < p_{\mathrm{T}} < 50$ GeV. These values are before the unfolding procedure is performed.

j_T [GeV]	dN/dj_T [GeV]$^{-1}$	(stat)
0-0.05	1.56e+03	216
0.05-0.1	4.37e+03	379
0.1-0.15	5.74e+03	427
0.15-0.2	6.46e+03	455
0.2-0.25	7.48e+03	510
0.25-0.3	7.11e+03	498
0.3-0.4	5.9e+03	331
0.4-0.5	4.98e+03	311
0.5-0.625	3.03e+03	220
0.625-0.75	2.44e+03	206
0.75-1	1.09e+03	97.3
1-1.25	561	70.9
1.25-1.5	162	38
1.5-2	106	23.1
2-3	14.2	5.9

Table 19: The efficiency corrected charged hadron-in-jet distributions for jet p_{T} $30 < p_{\mathrm{T}} < 50$ GeV. These values are before the unfolding procedure is performed.

r	dN/dr	(stat)
0-0.05	1.01e+04	651
0-0.05	1.25e+04	770
0.05-0.1	1.61e+04	831
0.1-0.15	1.16e+04	661
0.15-0.2	1.02e+04	597
0.2-0.25	7.72e+03	499
0.25-0.3	5.86e+03	429
0.3-0.35	4.58e+03	373
0.35-0.4	4.59e+03	374
0.4-0.45	3.4e+03	320
0.45-0.5	1.91e+03	236

Table 20: The efficiency corrected charged hadron-in-jet distributions for jet p_{T} $50 < p_{\mathrm{T}} < 100$ GeV. These values are before the unfolding procedure is performed.

z	dN/dz	(stat)
0.004-0.006	3.32e+03	1.49e+03
0.006-0.008	1.32e+04	3.02e+03
0.008-0.01	2.9e+04	4.53e+03
0.01-0.015	2.97e+04	2.93e+03
0.015-0.02	2.79e+04	2.88e+03
0.02-0.03	2.01e+04	1.77e+03
0.03-0.04	1.65e+04	1.66e+03
0.04-0.05	1.13e+04	1.41e+03
0.05-0.075	8.63e+03	822
0.075-0.1	5.99e+03	723
0.1-0.15	4.11e+03	443
0.15-0.2	3.14e+03	413
0.2-0.25	1.91e+03	335
0.25-0.5	499	78.2
0.5-0.75	55	28.6

Table 21: The efficiency corrected charged hadron-in-jet distributions for jet p_{T} $50 < p_{\mathrm{T}} < 100$ GeV. These values are before the unfolding procedure is performed.

j_T [GeV]	dN/dj_T [GeV]$^{-1}$	(stat)
0-0.05	601	141
0.05-0.1	2.06e+03	265
0.1-0.15	2.62e+03	296
0.15-0.2	2.82e+03	312
0.2-0.25	3.44e+03	366
0.25-0.3	3.09e+03	336
0.3-0.4	2.54e+03	222
0.4-0.5	2.07e+03	205
0.5-0.625	1.38e+03	159
0.625-0.75	1.04e+03	138
0.75-1	591	76.2
1-1.25	281	51.5
1.25-1.5	141	39
1.5-2	93.8	22.4
2-3	23.4	7.86

Table 22: The efficiency corrected charged hadron-in-jet distributions for jet p_{T} $50 < p_{\mathrm{T}} < 100$ GeV. These values are before the unfolding procedure is performed.

r	dN/dr	(stat)
0-0.05	8.92e+03	679
0.05-0.1	7.62e+03	579
0.1-0.15	4.87e+03	432
0.15-0.2	4e+03	379
0.2-0.25	3.12e+03	332
0.25-0.3	2.29e+03	272
0.3-0.35	1.89e+03	249
0.35-0.4	1.63e+03	224
0.4-0.45	1.39e+03	206
0.45-0.5	675	141